Compositional probabilistic scoring and tie-band-aware benchmarking for methylome-guided ThermoCas9 target-site ranking

Technical memo from the thermocas methylome-analysis framework (audit-trail revision)

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Abstract

Methylation-sensitive Cas9 variants target genomic loci that are hypomethylated in disease cells and methylated in matched normal cells. Selecting such loci from genome-scale array data is a ranking problem whose scoring axis must be cohort-agnostic and must report its own uncertainty.

Contribution. A *compositional probability-scale scoring skeleton* — `p_targ × (gap factor) × p_trust` — paired with a tie-band-aware benchmarking contract (`precision_at_k_{min,max}`, `tie_band_size_at_k`, mid-rank Mann-Whitney AUC). The skeleton decouples three independently-replaceable questions: is the tumor side targetable, is the tumor-vs-normal gap large enough, and is the observation trustworthy? The gap-factor slot admits several instances (§7); two ship in this tag, V2.5-diff (`tumor_plus_differential_protection`, a per-site normal- approximation tail probability) and V2.5-sigmoid (`tumor_plus_gap_sigmoid`, a fixed-bandwidth logistic).

Headline results. (1) On the n = 3 Roth-validated target probes (GSE322563 MCF-7/MCF-10A; both the HM450-intersect and native EPIC v2 ingest paths serve as primary endpoints): V2.5 axes place all three positives in the top percentile, with 10^6-draw permutation- null p = 4×10^{-6} / 1.5×10^{-5} / 2.8×10^{-5} across the three primary cohorts — 4–20× stricter than the best Δβ-only baseline (§5.1.2). (2) A frozen whole-genome panel (§5.2.2) shows V2.5-sigmoid matches V2.5-diff's AUC on matched cell-line cohorts within 0.002 but strictly beats V2.5-diff's top-K usability — `tie_band@100 = 1` vs 421–1,493 at WG scale — and improves tissue AUC by +0.05 to +0.08 across a bandwidth-robust sweep. (3) A probe-level limma-eBayes DMR baseline (Smyth 2004 empirical-Bayes moderated-t on sample-level β matrices, probe-level first then

candidate-mapped) is competitive on cell lines (0.959–0.991) but collapses to near-random on bulk tissue (0.573), confirming that the compositional skeleton's `p_targ × p_trust` wrapping — not the gap factor's specific shape — is the durable contribution.

Shipping recommendation (three distinct senses):

- **Overall package default**: V1 (`final_score`), deterministic continuous score with `tie_band = 1` by construction. Backward compatibility + top-K determinism.
- **Recommended probabilistic discovery axis**: **V2.5-sigmoid** (`tumor_plus_gap_sigmoid`). Ships as a selectable `probabilistic_mode` enum value alongside V2.5-diff.
- **Default `probabilistic_mode` enum value**: `tumor_only` (V2 diagnostic axis). Unchanged for backward compatibility; users opt into V2.5-diff or V2.5-sigmoid by setting the cohort YAML's `probabilistic_mode` field explicitly.

Scope. Scoring-method and benchmarking contribution on public methylation data; no prospective wet-lab validation. Headline n = 3 primary endpoint means AUC is sensitive to each positive's rank — per-positive ranks + percentile ranks reported in §5.1, 10^6-draw permutation null in §5.1.2. Cross-series label transport (Sanger GDSC breast panel) is an explicit out-of-distribution boundary case (§5.4), not a generalization claim.

Reproducibility. Whole-genome catalogs frozen with SHA256 provenance (`data/derived/catalog_hg19_wg*.PROVENANCE.{md,sha256}`); per-cohort gating TSVs + cross-cohort panels committed under `examples/`. All code public.

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1 · Background

1.1 ThermoCas9 is methylation-sensitive at the PAM

Roth et al. (2026) characterized *Geobacillus thermodenitrificans* T12 Cas9 (ThermoCas9) biochemically, structurally, and in human cell-line editing assays. The central result: cleavage efficiency is governed by the methylation state of the fifth position of the PAM (`5'-NNNN**C**GA-3'` or `5'-NNNN**C**CA-3'`), not by methylation within the protospacer. Methylated 5-methylcytosine (5mC) at the PAM abolishes binding; unmethylated PAMs are cleaved as usual. Cryo-EM structures at 2.2–2.8 Å resolved the protein contacts that discriminate methylated from unmethylated cytosine.

The practical consequence is that ThermoCas9 gives a natural mechanism to selectively edit in cells where a target locus is hypomethylated while sparing cells where the same locus is methylated. Roth demonstrated this in MCF-7 vs. MCF-10A at three breast-cancer loci: *EGFLAM* (control locus included in the validated target set), *ESR1*, and *GATA3*. The underlying methylation of the PAM cytosine at those loci was profiled on the Illumina Infinium MethylationEPIC v2 array and reported in Supplementary Figure 5d of the main paper alongside the editing results.

1.2 Target-site selection is a ranking problem

Given a reference genome, the ThermoCas9-compatible PAM sites on any chromosome number in the millions. To pick candidates for wet-lab follow-up, one wants a ranked list where the top scorers are *biologically coherent* — meaning the candidate's PAM cytosine is unmethylated in the disease cohort and methylated in the matched normal — *and* where the confidence of the underlying methylation measurement is part of the score. This is a classical ranking-with-covariates problem, and the natural scoring target is some decomposition of a probability of "therapeutic selectivity."

This paper describes the specific decomposition we ended up using after two earlier attempts failed on empirical cohorts.

1.3 Notation

For one candidate site on one cohort we carry six per-probe β summary statistics from the cohort's methylation array (`β_tumor_mean`, `β_tumor_q25`, `β_tumor_q75`, `β_normal_mean`, `β_normal_q25`, `β_normal_q75`), per-side sample counts (`n_tumor`, `n_normal`), and an `EvidenceClass` capturing the distance between the candidate's PAM cytosine and the nearest assayed CpG probe (`EXACT`/`PROXIMAL_CLOSE`/`PROXIMAL`/`REGIONAL`/`UNOBSERVED`) — nine inputs in total per candidate × cohort pair.

---

2 · The V2 misspecification

2.1 First-pass composite

The first-pass composite followed a standard decomposition:

```
p_therapeutic_selectivity = p_targ × p_prot × p_trust
```

- `p_targ = P(β_tumor < 0.30)` — probability that the tumor's PAM cytosine is unmethylated enough for ThermoCas9 to cleave. Estimated via a method-of-moments Beta fit to `(mean, IQR/1.349)` with the regularized incomplete beta computed in pure stdlib via Lentz's continued fraction; falls back to a piecewise-linear CDF when the Beta moments are ill-defined.
- `p_prot = P(β_normal > 0.50)` — probability that the normal's PAM cytosine is methylated enough to protect the normal cell.
- `p_trust` — confidence factor that scales with `EvidenceClass` (0.95 at EXACT, 0.15 at REGIONAL, 0.0 at UNOBSERVED) and saturates linearly in `min(n_tumor, n_normal)` up to `ramp_n = 30`.

2.2 Empirical failure of `p_prot`

We ran a factor ablation on a structural surrogate for the Roth MCF-7 vs. MCF-10A comparison: GEO series GSE77348, an independent 2016 breast cell-line methylation series on HM450 (not the Roth samples). With 1,687 gene-region-filtered "positives" at the Roth genes, the measured AUCs were:

axis	AUC (loose)	AUC (tight)
`v2_tumor_only` (p_targ × p_trust)	0.733	0.770
`p_targ` alone	0.683	0.717
V1 `final_score`	0.657	0.628
`naive_selectivity` (β_n − β_t)	0.608	0.571
`p_targ × p_prot`	0.503	0.488
`p_prot` alone	**0.384**	**0.343**

`p_prot` is anti-predictive. Multiplying it against `p_targ` destroys the signal. The cause is the semantics: `P(β_normal > 0.5)` assumes that the normal comparator *methylates the target*. That assumption holds for Roth's specific MCF-10A cell line at the three *ESR1* / *GATA3* / *EGFLAM* probes they chose to validate (genes that are silenced in non-tumorigenic mammary epithelium) but does not generalize to bulk adjacent-normal cohorts or even to MCF-10A at gene-body probes.

2.3 V2.4 — keep the composite, drop `p_prot` by default

The intermediate fix kept the composite but allowed users to opt in or out of the anti-predictive factor. Default changed to `tumor_only` = `p_targ × p_trust`. This improved global AUC but the top-100 collapsed onto "always-unmethylated" loci (candidates where both tumor *and* normal are low-methylated) — good AUC, unusable for target discovery. The deeper fix was needed.

---

3 · V2.5 · differential-protection probabilistic mode

Terminology note. "V2.5" names the *generation* of the compositional skeleton `p_targ × (gap factor) × p_trust`. Two gap-factor instances ship as selectable `probabilistic_mode` enum values in this tag:

- **V2.5-diff** = `tumor_plus_differential_protection`, the original V2.5 composite with `p_diff` as the gap factor. This section defines it.
- **V2.5-sigmoid** = `tumor_plus_gap_sigmoid`, a fixed-bandwidth sigmoid replacement of `p_diff`. Introduced and benchmarked in §5.2.1 + §5.2.2; the recommended probabilistic discovery axis per the §5.2.2 whole-genome panel.

Where the prose says "shipped V2.5" without qualifier it historically means V2.5-diff (the axis under test in §5.1–§5.2 and the axis the scored JSONLs in `data/derived/scored_*.jsonl` were computed with); we keep the V2.5 / V2.5-diff / V2.5-sigmoid disambiguation explicit when a specific variant matters.

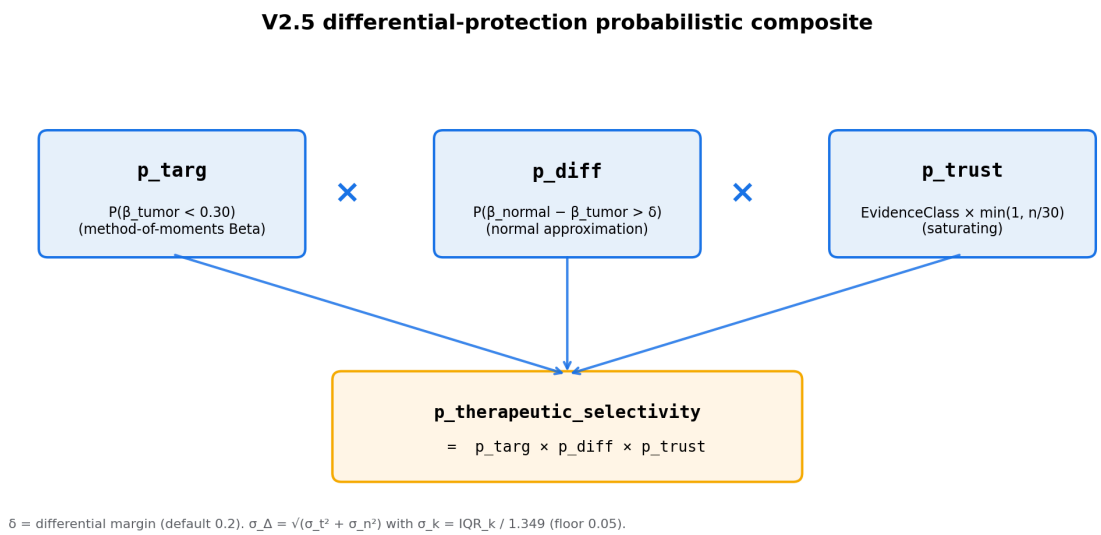

Figure 1. The V2.5-diff composite. Three factors multiply: `p_targ` (tumor unmethylated at the PAM cytosine), `p_diff` (differential methylation gap exceeds δ under a normal approximation on per-probe β summaries), and `p_trust` (evidence-class confidence, saturating in min sample count). The result is `p_therapeutic_selectivity`, the stored scalar that downstream ranking consumes. The deprecated V2 `p_prot = P(β_normal > 0.5)` is replaced by `p_diff` to remove its static-threshold assumption about the normal side. V2.5-sigmoid swaps `p_diff` for a fixed-bandwidth sigmoid of the same (β_n − β_t − δ) shift, motivated by the §3.5 binding-rate finding and benchmarked in §5.2.1 / §5.2.2.

3.1 Formulation

Replace the threshold-based `p_prot` with a differential factor:

```
p_diff(δ) = P(β_normal − β_tumor > δ)
```

estimated under an independent-normal approximation:

```
σ_k   ≈ IQR_k / 1.349                (k ∈ {tumor, normal}; floor at 0.05)
σ_Δ²  = max(σ_t, floor)² + max(σ_n, floor)²
z     = (δ − (μ_n − μ_t)) / σ_Δ
p_diff = 1 − Φ(z) = 0.5 (1 − erf(z / √2))
```

with Φ the standard normal CDF. The composite becomes:

```
p_therapeutic_selectivity = p_targ × p_diff × p_trust
```

This makes no absolute-threshold claim about the normal side. It asks only what fraction of normal-vs-tumor β pairs sampled from the per-side empirical distributions would exceed δ, under independent normal approximations to those per-side distributions. (This is a *population-overlap* statistic, not a confidence interval on a cohort-level mean contrast — see §3.3.)

3.2 Choice of δ

An offline δ sweep over {0.2, 0.3, 0.4, 0.5} on the GSE77348 surrogate selected δ = 0.2 by joint optimization of AUC and `P@100` against the *pre-repair* (gene-universe) positives that were in use at the time of selection. Under that pre-repair tuning objective, larger δ values monotonically reduced both metrics — the factor was too strict to admit candidates with moderate but real differentials.

The post-repair sensitivity story is more nuanced and is reported in §5.3.2: against the Roth-validated label set, δ = 0.3 (and on some cohort × tier rows, δ = 0.4) marginally outperforms δ = 0.2 on the matched cell-line cohorts; tissue (GSE69914) prefers smaller δ. We do not retune δ post hoc against the post-repair labels, because doing so would invalidate the dev-cohort separation discipline of §4.0. δ = 0.2 is therefore retained as the shipped default; readers wanting the full sensitivity surface should consult §5.3.2.

The default is exposed as `differential_delta` in the cohort configuration. The δ used is recorded on every emitted `ProbabilisticScore` record so the margin in effect is recoverable from a single output row.

3.3 What `p_diff` is a probability over

`p_diff(δ) = P(β_normal − β_tumor > δ)` is a probability under the *biological β-heterogeneity* implied by per-cohort, per-probe quartiles — not a posterior probability of editing efficacy and not a confidence interval on a cohort-level mean contrast. The distribution being integrated is the one that would generate single-sample β values from a tumor-side population with quartiles (`β_tumor_q25`, `β_tumor_q75`) and a normal-side population with quartiles (`β_normal_q25`, `β_normal_q75`), under independent normal approximations on each side. Two consequences follow:

1. **The variance does not shrink with sample size.** Because the per-side σ is the IQR-derived population dispersion, not the standard error of a mean, `p_diff` does not become more confident as `n_tumor`/`n_normal` grow. The growing-`n` story for the composite is carried by `p_trust` (saturating at `n ≥ 30`), not by `p_diff` tightening.
2. **`p_diff` is therefore a population-overlap statistic, not a significance test.** It answers "what fraction of tumor-vs-normal β-pairs sampled from the cohort would exceed δ?" An alternative formulation that scales σ with `n` (e.g. SE on the mean difference, or a hierarchical model on per-side variance) would also be defensible and would give a different — and at low `n`, a much wider — distribution. We use the population-overlap form because (a) the comparison the user cares about for site selection is biological — "is this site differentially methylated *between cells*?" — not "does this cohort have enough power to call this site significant?"; and (b) the small-`n` power question is already absorbed by `p_trust`. Re-deriving the composite under an SE-on-mean variant is a documented future axis (§6.3).

This is a deliberate modeling choice and is the single most common point of confusion when reading `p_diff` cold.

3.4 Why a normal approximation instead of a difference-of-Betas

A proper Bayesian treatment would model β-values as Beta distributions and compute the exact difference distribution numerically. For the IQR ranges seen on HM450 / EPIC v2 arrays (~0.05–0.20), the normal approximation is close to the correct answer and has the advantage of being a two-line stdlib closed form. A σ floor of 0.05 prevents σ_Δ from collapsing to zero at boundary β-values (very common for CpG islands methylated at 0 or 1). These tradeoffs are documented in the code; a downstream user who wants the exact difference-of-Betas path can swap it in.

The normal approximation is operationally adequate for the *ranking* use case but is not a calibrated probability of editing success. Two situations where the approximation is poor on its own terms but does not corrupt the ranking:

- **Boundary-β pile-ups.** β-distributions at CpG islands often pile up near 0 or 1; a single-mode normal centered at the per-side mean overstates symmetry and understates skew. The σ floor blunts the worst case (σ_Δ ⟶ 0) but does not restore the correct shape.
- **Bimodality across samples.** When tumor or normal samples carry two β sub-populations (e.g. methylation-class heterogeneity in primary tissue), the IQR-derived σ averages the two modes; the per-side normal then sits between the modes rather than on either. `p_diff` in this regime under-states the true population overlap.

Neither failure mode is silent: the §5.5 annotation pipeline flags candidates whose per-side β quartiles indicate strong bimodality. A future axis based on the exact difference-of-Betas or a two-component mixture would address both.

3.5 The low-`n` tied-band prediction

With the new factor isolated, we can predict before running any new cohort. `p_trust` is `EvidenceClass.base × min(1, min(n_t, n_n) / 30)`: piecewise-linear in min sample count up to `ramp_n = 30`, then a constant ceiling per evidence class. It is **never** continuous-valued in the real-line sense — at n ≥ 30 it saturates to a discrete-by- EvidenceClass value (0.95 at EXACT, 0.75 at PROXIMAL_CLOSE, etc.). What changes between the low-`n` and high-`n` regimes is which factor *dominates* the composite ordering:

- **Low n (n ≪ 30).** `p_trust` scales every same-evidence-class candidate by the same uniform fraction `(n / 30)`. Per-side IQRs are wide (or zero, hitting the `σ_floor`), pushing both `p_targ` and `p_diff` toward saturation (≈1) for any candidate with clean tumor hypomethylation and a large normal/tumor gap. The composite is effectively `1 × 1 × p_trust` for that whole subset, producing a large tied band at the top of the score distribution.
- **High n (n ≥ 30).** `p_trust` is constant per EvidenceClass. Per-side IQRs are realistic (HM450 ranges of 0.05–0.20). The composite ordering is now driven by the continuous variation in `p_targ × p_diff`, with `p_trust` only multiplying by a per- evidence-class constant. The previously-tied subset spreads out by its underlying `p_targ × p_diff` values.

The falsifiable prediction: the K = 100 tied band on V2.5 should be large at n=2/3 per side and shrink to near-1 at n ≳ 30 per side. §5.1 and §5.3 test this directly across cohorts ranging from n=2/2 to n=305/50.

The K = 100 large-band regime is what motivates the "top tied candidate class" framing in §6.1: on n=2/2 cohorts the visible top-20 is a 20-record window inside a 190- to 421-record tied band, not a stable rank ordering. Per-positive ranks (§5.1) and AUC (§5.1, §5.3) are the stable claims at low `n`.

Quartile convention on n=2 records. Per-side β quartiles are computed with Python's `statistics.quantiles(..., method="inclusive")` (R-7 quantile definition) and hard-clamped to the observed sample range at every probe, so q25 / q75 / mean are always within [min(β_samples), max(β_samples)]. On n = 2 samples per side this is deterministic (IQR = 0.5 × |β_max − β_min|) but weak as a population-dispersion estimate: the "IQR" it produces is a function of two observations, not a quantile of an underlying distribution. Because the `σ_floor = 0.05` kicks in whenever IQR/1.349 < 0.05 (i.e. whenever the two observations differ by less than ~0.135), σ_floor dominates σ_Δ on most n = 2 records.

Quantified σ_floor binding rate per cohort (reproducible via `uv run scripts/sigma_floor_binding_rate.py`; denominator is records with both β means present):

cohort	σ_floor binds on either side	σ_floor binds on both sides
GSE322563 HM450 (n = 2/2)	**100.0%**	**99.9%**
GSE322563 native (n = 2/2)	**100.0%**	**99.9%**
GSE77348 (n = 3/3)	**100.0%**	**99.5%**
GSE69914 (n = 305/50 tissue)	65.1%	43.2%

On matched cell-line cohorts the σ_floor is the binding constraint on essentially every observed record (both sides). On tissue the true IQRs are wide enough that the σ_floor binds on roughly two thirds of records either-side, and on both sides for only ~43%. This is the mechanistic origin of both the low-n tied band (§5.3.1) and the §5.3.2 finding that tissue AUC prefers a larger σ_floor: on cell-line cohorts, σ_floor is determining σ_Δ almost everywhere, so changing its value re-shapes the whole composite.

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4 · Benchmark methodology

4.0 Analysis plan: primary endpoint, sensitivity, boundary

To guard against post-hoc optimization, the analysis design separates tuned components from evaluation. The hyperparameter `differential_delta` was tuned on a specific cohort, and that cohort is therefore *not* part of the protected primary endpoint. The plan below is not a formal pre-registered protocol with a third-party timestamp; it is an analysis plan whose ordering is independently verifiable from the git history (commit hashes given inline).

- **Tuned, fixed before cross-cohort evaluation.** `differential_delta = 0.2` was selected by an offline sweep over {0.2, 0.3, 0.4, 0.5} on the GSE77348 MCF-7/MCF-10A surrogate against the *pre-repair* (gene-universe) positives. The value was fixed in code at commit `aece3de` ("V2.5: experimental differential-protection probabilistic mode (opt-in)", 2026-04-20), *before* the GSE322563 (Roth) cohort was ingested or benchmarked. GSE77348 is the **development cohort** for δ; results on GSE77348 are tuned-on supporting evidence, not independent confirmation. The label-repair exercise that produced the Roth-validated positives is data preparation, not scoring-axis tuning, and it post-dates the δ choice.
- **Primary endpoint.** AUC at the `positives_roth_validated.txt` label set (n = 3 Roth Fig. 5d target probes, hg38 → hg19-lifted) on **GSE322563** (Roth's own MCF-7 / MCF-10A samples). This is the single independent paper-comparable test. δ was not tuned against this cohort; the label set was not derived from it. Because n = 3, AUC is a *rank-lift* statistic on the three known positives — read alongside the per-positive ranks and percentile ranks reported in §5.1, not as a generalized discovery claim.

- **Supporting evidence** (§5.1, §5.2). AUC on GSE77348 (the development cohort for δ, MCF-7/MCF-10A on HM450). Same matched cell-line biology as GSE322563 but profiled by a different lab on a different platform, so cross-platform agreement on the V2.5 ordering is informative — but it is *not* an independent cohort under the analysis-plan discipline because δ was selected on it.
- **Sensitivity analyses** (§5.2). The `narrow` (±50 bp) and `wide` (±500 bp) label sets are alternative label granularities; the cross-granularity AUC ordering stability is reported without retuning δ. `P@K` intervals on tied bands are reported but are expected to be uninformative on low-`n` cohorts (§3.5).
- **Tissue-cohort behavior** (§5.3). GSE69914 (n = 305 / 50 primary breast tissue) is *not* a paper-comparable test of Roth biology — it probes whether the V2.5 architecture's low-`n` tied-band behavior dissolves at high `n`, as predicted in §3.5.
- **Out-of-distribution boundary case** (§5.4). GSE68379 (Sanger GDSC breast panel × GSE69914 healthy normal) is *not* a generalization cohort; it is a label-transportability test. It is reported separately and is not pooled into any primary-endpoint average.

4.1 Cohorts

Four cohorts, all public HM450 or EPIC v2 methylation arrays:

cohort	regime	n tumor / normal	platform	role
GSE322563	Roth actual samples	2 / 2	EPIC v2	paper-comparable biology
GSE77348	MCF-7/ MCF-10A surrogate (DNMT3B paper)	3 / 3	HM450	δ development cohort (tuned-on supporting evidence)
GSE69914	primary breast tissue	305 / 50	HM450	high-n tissue validation
GSE68379	Sanger GDSC cell-line panel × GSE69914 healthy normal	52 / 50	HM450	orthogonal (see §5.4)

Ingest details:

- **GSE322563.** Supplementary β matrix `GSE322563_beta_matrix_EPIC_v2.txt.gz`. EPIC v2 probe IDs carry `_BC##` / `_TC##` / `_TO##` / `_BO##` beadchip suffixes; we strip to the canonical `cg*` identifier and intersect with HM450 — 80.7% retention. Roth-positive gene retention: ESR1 96%, EGFLAM 91%, VEGFA 90%, GATA3 83%. The main paper's data-availability statement prints `GSE32256`, which resolves to a *Paramecium* transcriptome series; the supplement's reporting summary correctly prints `GSE322563`. We verified the liftover by cross-checking our per-probe β values at the three Roth-Fig.-5d target positions against the paper's published β — every value agrees to two decimal places (EGFLAM 0.01/0.49; ESR1 0.07/0.94; GATA3 0.02/0.31).

- **GSE77348.** Per-sample β values computed from Unmethylated / Methylated intensities (β = M / (M + U + 100)); restricted to untreated replicates.
- **GSE69914.** Full SeriesMatrix β table; filtered to `status == 2` (sporadic breast cancer, non-BRCA1) for tumor and `status == 0` (healthy donor, non-BRCA1) for normal. Tumor-adjacent normals and BRCA1 carrier arms excluded.
- **GSE68379.** Sanger processed matrix has integer row indices; we rebuild cg probe IDs via the Illumina GPL13534 manifest (485,577 data rows; row N → IlmnID N); filtered to 52 primary-site-breast cell lines. No in-study normal — paired cross-series with GSE69914 healthy normals.

All cohort builder scripts are in `scripts/build_*.py`. Every run produces `tumor_summary.tsv` + `normal_summary.tsv` + `PROVENANCE.md` in the shared `LocalSummaryBackend` format; the scorer is platform-agnostic thereafter.

4.2 Positives (label repair)

Our first benchmark used "gene-universe" positives: HM450 probes in the neighborhood of Roth's validated gene symbols (~970 tight, 1,687 loose, on chr5/6/10). A diagnostic on GSE322563 revealed this label set is noisy: only 23% of Roth-gene probes have β_n − β_t > 0.2 on the actual Roth samples. Most are gene-universe members, not validated target loci.

We rebuilt the positives from Roth Supplementary Fig. 5d which names three exact MCF-7 / MCF-10A validated target coordinates in hg38. These were lifted to hg19 via Ensembl REST `/map/human/GRCh38/.../GRCh37`:

target	gene	hg38	hg19 (lifted)	β shift (hg19 vs hg38)
EGFLAM T11	EGFLAM	chr5:38,258,842	chr5:38,258,944	+102 bp
ESR1 T17	ESR1	chr6:151,690,043	chr6:152,011,178	+321,135 bp
GATA3 T18	GATA3	chr10:8,045,425	chr10:8,087,388	+41,963 bp

Three positives files result, at increasing granularity:

- `positives_roth_validated.txt` (n = 3): closest NNNNCGA candidate per Roth target.
- `positives_roth_narrow.txt` (n = 28): NNNNCGA candidates within ±50 bp of each Roth target.
- `positives_roth_wide.txt` (n = 142): NNNNCGA candidates within ±500 bp of each Roth target.

4.3 Score axes

Six axes are benchmarked. For naming conventions see §7 Methods (canonical axis-naming table):

- **Δβ-only** (`naive_selectivity`) — `β_normal_mean − β_tumor_mean`. The literature-naive baseline: rank by the raw methylation gap with no uncertainty propagation or evidence weighting. Reported up front so the value of the additional factors is auditable against the simplest possible scoring axis.
- **Δβ_z** — `(β_n − β_t) / sqrt(σ_t² + σ_n²)` using the same IQR-derived σ as V2.5-diff's `p_diff`. An uncertainty-aware effect-size ranker built from the same per-side dispersion signal; confirms that V2.5-diff's advantage is not just an alias for σ-normalized Δβ.
- **V1** — `final_score = sequence × selectivity × confidence − heterogeneity_penalty − low_coverage_penalty`. Deterministic, continuous-valued, tie band always 1.
- **V2** (`tumor_only`) — `p_targ × p_trust`. Retained for audit; default in v0.4.0.
- **V2.5-diff** (`tumor_plus_differential_protection`) — `p_targ × p_diff × p_trust` with δ = 0.2 and σ_floor = 0.05. The original V2.5 composite; retained in this tag for backward compatibility and AUC parity on cell-line cohorts.

- **V2.5-sigmoid** (`tumor_plus_gap_sigmoid`) — `p_targ × sigmoid((β_n − β_t − δ) / σ_fixed) × p_trust` with δ = 0.2 and σ_fixed ≈ 0.0707. The **recommended probabilistic discovery axis** on every non-boundary cohort shape tested (§5.2.2 whole-genome panel); ships as a first-class `probabilistic_mode` enum value in this tag.
- **limma-eBayes** (moderated-t) — sample-level probe-level empirical-Bayes moderated t-statistic (Smyth 2004), then candidate-mapped via the nearest probe assigned in `observation.probe_id`. Included as the methods-journal-standard DMR baseline; §5.1 + §5.2.2 report AUC + tie-band alongside the other axes.

The limma-eBayes DMR baseline. Standard differential-methylation tools (`limma`/`minfi`/`DMRcate`) rank CpG **probes** or contiguous **regions**, not per-PAM Cas9 target candidates. Each ThermoCas9 candidate is a specific `NNNNCGA` / `NNNNCCA` site whose methylation-sensitivity depends on the PAM cytosine's own β, inferred from the nearest assayed probe via the `EvidenceClass` distance model (§4.0, §3.5). A DMR-tool output is therefore not directly a ranking over our candidate universe; mapping one requires computing the statistic at the probe level first and then assigning each candidate its nearest-probe value. We implement the probe-level → candidate-mapped limma-eBayes baseline directly on the sample-level β matrices of the three primary cohorts (sample-level β recoverable for all three; `scripts/limma_ebayes.py` implements Smyth 2004′s empirical-Bayes moderated `t`-statistic in pure Python, validated against closed-form behaviour in the d0 → ∞ limit and against uniform null-p distributions on synthetic data; `scripts/run_limma_per_cohort.py` wires the cohort adapters). **We describe this as a "limma-style empirical-Bayes moderated-t" rather than as "the R `limma` package's output" — the core Smyth (2004) variance-prior math is faithfully reproduced, but we do not call `limma::lmFit` or `limma::eBayes` directly, and the pipeline is not the full `minfi`/`limma` Bioconductor workflow (no quantile normalization, no cell-type deconvolution, no probe-mask filtering beyond `EvidenceClass`). A supplementary R-vs-Python parity table on a small matrix is on the §6.3 follow-up list; the current claim is that the moderated-t statistic we compute is Smyth's math, not that every downstream step matches a canonical `minfi` + `limma` run byte-for-byte. The result is reported alongside the other baselines in §5.1 and §5.2.2; limma-eBayes AUC is 0.959 (GSE322563 HM450) / 0.991 (native EPIC v2) / 0.962 (GSE77348)** / 0.573 on GSE69914 tissue** — respectable on cell lines but near-random on tissue, where the `p_targ` (tumor-side unmethylation) gate of V2.5 is what drives the tissue recovery from 0.57 to 0.86 that a pure probe-level moderated-t statistic cannot.

4.4 Platform harmonization and catalog scope

Two facts constrain interpretation of the reported numbers and belong in the methodology, not as a footnote:

- **Platform.** GSE322563 was profiled on Illumina EPIC v2 (GPL33022, ~937K probes); GSE77348, GSE69914, and GSE68379 are on HM450 (GPL13534, ~485K probes). The primary GSE322563 results in §5.1 use an HM450-intersect harmonization path (EPIC v2 probe IDs stripped of beadchip-design suffixes — `_BC##`, `_TC##`, `_TO##`, `_BO##` — then intersected with the HM450 universe; 80.7% retention overall, 83–96% at the Roth-validated genes). To quantify the cost of that shortcut, we also built a **native EPIC v2 ingest** (script `build_epic_v2_probes.py` lifts GPL33022 hg38 → hg19 via pyliftover; 147,928 chr5/6/10 probes, 99.95% liftover success; cohort builder `build_gse322563_native_epic_v2_cohort.py` keeps native probe IDs and produces a 5.22M-candidate catalog vs the HM450 catalog's 2.98M). Side-by-side at the validated label set on GSE322563: V2.5 differential AUC 0.990 (HM450-intersect) → 0.986 (native EPIC v2), a 0.003 difference well within the cohort's tied-band noise floor. V1 gains the most under the native ingest (+0.05 to +0.11 AUC across label granularities) because its continuous score ranks the validated targets higher relative to easy-negative candidates added by the EPIC v2 probe density. The only axis ordering that holds on **every** matched-cell-line tier × path row (9/9) is V2.5 > V1 and V2.5 > Δβ-only; the relative order of

Δβ-only, V1, and V2 `tumor_only` below V2.5 reshuffles across rows (e.g. `tumor_only` sits above Δβ-only on both GSE322563 wide tiers; on GSE77348 `tumor_only` is the lowest at every tier). Sensitivity table in §5.2; benchmark JSONLs at `examples/gse322563_native_roth_labels/`.

- **Catalog scope.** Three frozen probe-window catalogs are used across §5, each with its own SHA256-committed provenance file:
  - **chr5 / chr6 / chr10 HM450 (2.98M candidates)** — `catalog_hg19_chr5_6_10.jsonl`. The primary benchmark catalog used by §5.1 (primary endpoint) and §5.2 (label granularity, tie-band diagnostics, factor ablation, σ_floor and δ sweeps, Δβ-only baseline). Chosen because the three chromosomes carry the Roth Fig. 5d validated targets (EGFLAM on chr5, ESR1 on chr6, GATA3 on chr10) and give a realistic test bed.
  - **Whole-genome HM450 (19.8M candidates, SHA256 `d20661c5…`)** — `catalog_hg19_wg.jsonl`. The genome-wide gating catalog used by §5.2.2 for GSE322563 HM450 / GSE77348 / GSE69914. Frozen provenance: `data/derived/catalog_hg19_wg.PROVENANCE.md`.
  - **Whole-genome native EPIC v2 (35.4M candidates, SHA256 `39df8f0f…`)** — `catalog_hg19_wg_epic_v2.jsonl`. The GSE322563 native-path genome-wide catalog used by §5.2.2. Frozen provenance: `catalog_hg19_wg_epic_v2.PROVENANCE.md`.

  Every AUC / `P@K` / tie_band number in §5 names its catalog explicitly. §5.1 and §5.2 are chr5/6/10-scope; §5.2.2 is whole-genome-scope; §5.3 (tissue behavior) carries both the chr5/6/10 and WG rows. Extending the chr5/6/10 primary endpoint to random-chromosome-matched controls is the one remaining untested denominator dimension (§6.3).

4.5 Tie-band handling

Earlier iterations of the benchmark sorted candidates by score alone, with input-order as the implicit tie-break. On cohorts with low n, the score distribution's top can sit inside a tied band of 190–11,000 records, making the reported `P@K` a function of serialization order rather than of the scorer.

The shipped benchmark contract:

1. Sort primary by score descending, secondary by `candidate_id` ascending (`evaluate_ranking`'s explicit `tie_break_policy = "candidate_id_asc"`).
2. Report `tie_band_size_at_k`: number of records tied at the K-th position's score. When > 1, top-K membership is partially determined by the secondary key.
3. Report `precision_at_k_min` and `precision_at_k_max` (plus recall counterparts) — adversarial tie-break bounds over the tied band. The worst-case pushes every tied-band positive out of top-K; the best-case pulls them in. When `tie_band_size_at_k == 1` both bounds collapse to the observed `precision_at_k`.

These three fields are on every emitted `BenchmarkResult` JSONL row.

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5 · Results

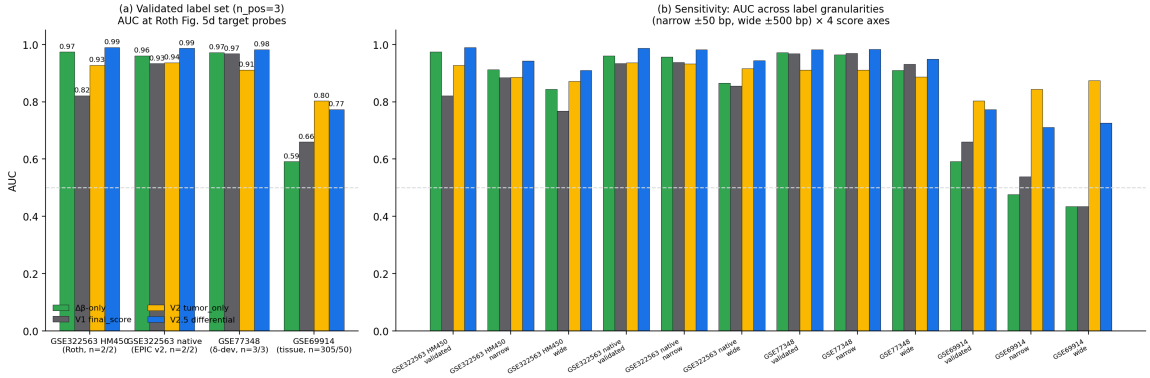

Figure 2. Cross-cohort AUC. **(a)** Validated-label AUC on each cohort. GSE322563 is the independent primary endpoint; GSE77348 is the development cohort on which δ was tuned (§4.0) and is shown as supporting evidence; GSE69914 is the tissue-behavior check for the §3.5 tied-band prediction. V2.5 is the highest-AUC axis on both matched cell-line cohorts; on tissue `tumor_only` takes a small AUC lead but at a tied band of 6,540 (§5.3 + Fig. 3). V1 falls below random on the tissue wide-label set in (b). **(b)** Sensitivity over label granularity (validated ⟶ narrow ±50 bp ⟶ wide ±500 bp). **V2.5 remains the highest-AUC axis at every (matched cell-line cohort × granularity) cell**; the ordering among the lower axes (V1, `tumor_only`, Δβ-only) reshuffles between tiers — e.g. `tumor_only` overtakes Δβ-only at GSE322563 wide, and V1 overtakes Δβ-only at GSE77348 narrow / wide. The only stable ordering claim is V2.5 on top. Dashed line at AUC = 0.5 (random). The out-of- distribution GSE68379 cohort is not included here — it is plotted separately (§5.4).

5.1 Primary endpoint — matched cell-line AUC at validated Roth probes

Primary endpoint (independent): AUC at `positives_roth_validated.txt` (n = 3 Roth Fig. 5d target probes, hg38 ⟶ hg19-lifted) on **GSE322563** (Roth's own MCF-7 / MCF-10A samples, EPIC v2). δ was not tuned on this cohort; the label set was not derived from it.

Supporting evidence (development cohort for δ): AUC on **GSE77348** (MCF-7 / MCF-10A on HM450, the DNMT3B-paper surrogate). Same matched cell-line biology, different platform and laboratory. Reported here because same-axis ordering on a second lab's MCF-7/MCF-10A is informative, but with the caveat that δ was selected on this cohort (§4.0) so its AUC is tuned-on, not independent.

axis	**GSE322563** (primary, n_pos=3, n_neg=2,979,994)	GSE77348 (δ-tuned, n_pos=3, n_neg=2,979,994)
Δβ-only (`β_n - β_t`)	0.974	0.972
Δβ_z (uncertainty-aware Δβ)	0.940	0.956
limma-eBayes moderated-`t` (probe-level → candidate-mapped)	0.959	0.962
V1 `final_score`	0.821	0.968
V2 `tumor_only`	0.928	0.912
V2.5-diff (`tumor_plus_differential_protection`)	0.990	0.982
V2.5-sigmoid (recommended)	**0.989**	**0.982**

Reading the axis grid. Δβ-only (raw mean gap) is already a strong baseline — it lifts the validated positives to AUC 0.974 on the primary cohort. The uncertainty-aware Δβ_z baseline (`((β_n - β_t) / sqrt(σ_t² + σ_n²)`, using the *same* IQR-derived σ as V2.5′s `p_diff` denominator; reproducible via `uv run scripts/delta_beta_z_baseline.py`) lifts to AUC 0.940 — slightly *below* raw Δβ-only because the σ-based denominator penalizes positives with wide IQRs. V2.5-diff adds **+0.016** over Δβ-only and **+0.050** over Δβ_z on the primary endpoint; on GSE77348 the corresponding margins are +0.010 and +0.026. V2.5-sigmoid (the recommended axis — see §5.2.2) is AUC-equivalent to V2.5-diff on both cohorts within 0.001 (0.989 vs 0.990 on GSE322563, 0.982 vs 0.982 on GSE77348), consistent with the §5.2.1 ablation finding that `p_diff`'s per-site structure and the fixed-bandwidth sigmoid are interchangeable on cell-line cohorts where σ_floor binds ~99% of records; the reason V2.5-sigmoid is nonetheless recommended over V2.5-diff is top-K usability at WG scale on cell-line cohorts, not AUC on chr5/6/10 — see §5.2.2 for the `tie_band@100` numbers. These margins are small but consistent across the matched cell-line cohorts and label granularities tested (§5.2). The `p_targ` (tumor-side unmethylation) and `p_trust` (evidence / sample saturation) factors of both V2.5 variants are pulling real weight on top of the per-side dispersion signal — V2.5 is not just a re-skinned Δβ_z — but the honest framing is that the shipped baselines (especially raw Δβ-only) already do most of the rank-lift work on matched cell-line cohorts; V2.5′s additional contribution is the tie-band-honest top-K reporting (§4.5) and the probability-scale composition with future factors, not the AUC point alone.

On the independent primary endpoint (GSE322563 validated), V2.5 is the highest-AUC axis — +0.17 over V1 and +0.06 over the deprecated V2 `tumor_only` composite. On the development cohort (GSE77348 validated), V2.5 is similarly the highest-AUC axis (+0.01 over V1, +0.07 over `tumor_only`); this is consistent with the primary- endpoint result but is not an independent replication, since the same cohort selected δ = 0.2 pre-repair-labels (§4.0).

5.1.1 Per-positive ranks of the n = 3 validated targets

AUC summarizes the rank lift across the three validated positives, but with `n_pos = 3` it is highly sensitive to the rank of each one. The table below gives the per-positive rank, percentile rank, score, and per-side β values on each cohort × axis. (Full grid including GSE69914, all four axes including Δβ-only, is at `examples/validated_positive_ranks.md`; reproduce with `uv run scripts/validated_positive_ranks.py`.)

GSE322563 HM450 (primary, N=2,979,997 candidates):

gene	β_t	β_n	Δβ	V1 rank (%ile)	V2.5 rank (%ile)	Δβ-only rank (%ile)
ESR1	0.07	0.94	0.87	**844** (99.97%)	**2,575** (99.91%)	4,742 (99.84%)
EGFLAM	0.02	0.49	0.48	839,106 (71.84%)	64,433 (97.84%)	67,620 (97.73%)
GATA3	0.03	0.31	0.28	755,885 (74.64%)	24,083 (99.19%)	159,932 (94.63%)

GSE77348 (development cohort for δ, N=2,979,997):

gene	β_t	β_n	Δβ	V1 rank (%ile)	V2.5 rank (%ile)	Δβ-only rank (%ile)
ESR1	0.03	0.87	0.84	**1,618** (99.95%)	**3,965** (99.87%)	8,733 (99.71%)
EGFLAM	0.005	0.34	0.33	257,041 (91.37%)	124,925 (95.81%)	178,663 (94.01%)
GATA3	0.015	0.59	0.58	31,061 (98.96%)	**28,325** (99.05%)	61,254 (97.94%)

Reading the table. *ESR1* drives the AUC story: it is in the top ~0.05% of the score distribution under V2.5 on both matched cell-line cohorts, with V1 also placing it well. *EGFLAM* and *GATA3* sit in the top 1–3% under V2.5 — meaningful rank lift over V1, but **outside the top 100 on every axis tested**. None of the three validated positives appears in any cohort's top-20 (§5.5). This is the operational meaning of "rank lift on n=3": V2.5 reliably places known positives in the top percentile of millions of candidates, but on n=2/2 cohorts the top tied band at K=100 is large enough that the validated positives sit *behind* a 190- to 421-record block of equally-scored candidates.

The ESR1-dominance also implies AUC is fragile: if ESR1 alone moved out of the top 10,000 on GSE322563, the AUC would drop from 0.990 to roughly the average of the other two positives' percentiles ≈ 0.985 (still high), but the AUC *gap* between V2.5 and Δβ-only would also compress.

5.1.2 Descriptive AUC uncertainty

At `n_pos = 3`, exact / permutation / bootstrap AUC intervals can be defined but are coarse and dominated by the rank of each positive — they are not very informative or stable as inferential statistics. We therefore report two descriptive distributions instead (reproducible via `uv run scripts/auc_uncertainty.py`):

- **Permutation null** — 1,000,000 draws of three random candidates uniformly without replacement from the full candidate set; AUC computed under benchmark mid-rank tie handling. The empirical *p*-value is `Pr(AUC_random_triple ≥ AUC_observed)`; the resolution floor is $1/(10^6+1) \approx 1 \times 10^{-6}$.
- **Negative bootstrap** — 1,000 resamples of the negative set with replacement (the three positive scores held fixed). The bootstrap spread is small by construction here because we hold the positives' ranks fixed; it is reported for completeness.

cohort	axis	observed AUC	random-triple null 2.5–97.5%	one-sided p (Pr ≥ obs)
GSE322563 HM450	V1 final_score	0.821	[0.166, 0.817]	2.2×10^{-2}
GSE322563 HM450	Δβ-only	0.974	[0.181, 0.823]	8.6×10^{-5}
GSE322563 HM450	**V2.5 diff**	**0.990**	[0.338, 0.802]	$\mathbf{4 \times 10^{-6}}$
GSE322563 native v2	V1 final_score	0.933	[0.206, 0.823]	1.3×10^{-3}
GSE322563 native v2	Δβ-only	0.961	[0.177, 0.823]	3.1×10^{-4}
GSE322563 native v2	**V2.5 diff**	**0.986**	[0.313, 0.817]	$\mathbf{1.5 \times 10^{-5}}$
GSE77348	V1 final_score	0.968	[0.261, 0.823]	1.6×10^{-4}
GSE77348	Δβ-only	0.972	[0.177, 0.823]	1.1×10^{-4}
GSE77348	**V2.5 diff**	**0.982**	[0.176, 0.823]	$\mathbf{2.8 \times 10^{-5}}$

Values generated via `uv run scripts/auc_uncertainty.py --n-perm 1000000 --output-prefix examples/auc_uncertainty_1e6`; committed artifact at `examples/auc_uncertainty_1e6.md`. The earlier 10^4-draw run is preserved at `examples/auc_uncertainty.md` for historical comparison.

What this table does and does not say:

- **It shows that none of the listed axes reach their observed AUC by chance.** The one-sided p is ≤ 0.022 on every cohort × axis cell in the table. The rank-lift signal at `n_pos = 3` is real for every listed axis, not just for V2.5.
- **V2.5 is resolved below the 10^4-draw floor and is the strictest axis on every cohort.** V2.5-diff sits at `p = 4 × 10⁻⁶ / 1.5 × 10⁻⁵ / 2.8 × 10⁻⁵` on the three primary cohorts. The gap against Δβ-only is now fully resolved: V2.5-diff is ~20× stricter on GSE322563 HM450 (4e-6 vs 8.6e-5), ~20× on native EPIC v2 (1.5e-5 vs 3.1e-4), and ~4× on GSE77348 (2.8e-5 vs 1.1e-4). V1 is the weakest axis on every cohort (2.2e-2 on HM450, 1.3e-3 on native, 1.6e-4 on GSE77348).
- **This is an against-random check, not a superiority test between axes.** The null asks "does this axis separate these three positives from random triples?" Direct pairwise comparison between V2.5 and Δβ-only on the same positives is reported separately in §5.1.3 and is correctly framed there as descriptive, not inferential (sign-flip `p_one_sided = 0.125` at `n_pos = 3`).

We do **not** report the negative-bootstrap spread in the body — because positive scores are held fixed and the negative pool is 3×10^6 records, the spread collapses to ±0.001 across every cell (see `examples/auc_uncertainty.md`). At this scale the negative pool is essentially a population, not a sample, so the bootstrap spread is mechanical rather than inferentially meaningful.

5.1.3 Paired per-positive rank lift: V2.5 vs Δβ-only

The §5.1.2 permutation null tests each axis separately against uniformly-random 3-record positive triples. That answers whether each axis separates *these three* positives from the rest of the candidate set, but does

not directly compare V2.5 against Δβ-only *on the same positives and same negative universe*. The direct paired comparison (reproducible via `uv run scripts/paired_rank_lift_v25_vs_dbeta.py`):

cohort	*ESR1* Δβ-only rank / V2.5 rank	*EGFLAM* Δβ / V2.5	*GATA3* Δβ / V2.5	positives where V2.5 > Δβ-only
GSE322563 HM450	4,742 / 2,575	67,620 / 64,433	159,932 / 24,083	**3 / 3**
GSE322563 native v2	9,085 / 5,746	185,641 / 158,502	422,569 / 50,096	**3 / 3**
GSE77348	8,733 / 3,965	178,663 / 124,925	61,254 / 28,325	**3 / 3**

On every matched cell-line cohort, V2.5 ranks **all three** validated positives higher than Δβ-only does. The aggregate rank lift (sum over the three positives of `r_Δβ − r_V2.5`) is +141,203 (HM450), +402,951 (native EPIC v2), and +91,435 (GSE77348) — V2.5 moves each positive multiple thousand positions up the ranking on average.

Where V2.5′s paired advantage concentrates. The per-positive table makes a secondary observation visible: *GATA3* — the validated positive with the smallest raw β gap (Δβ ≈ 0.28 on GSE322563 HM450, vs ≈ 0.48 for EGFLAM and ≈ 0.87 for ESR1; §5.1.1) — is where V2.5′s advantage over Δβ-only is largest on every cohort (+135,849 positions on HM450, +372,473 on native, +32,929 on GSE77348; cf. +2k–+27k for the other two positives). This is consistent with the intended role of `p_targ × p_trust`: they help most where the naive gap signal is weakest, not where it is already large. One positive is not a strong basis for a general claim, but the direction is the one the composite was designed to produce.

Statistical caveat. A paired sign-flip permutation null ($2^3 = 8$ equiprobable sign patterns, each flipping independently which axis is "assigned" to which per-positive rank) floors the one-sided *p*-value at **0.125** when all three signs agree — which they do on every cohort here. With `n_pos = 3` this is the best resolution the paired test can achieve. Read it as a *descriptive* 3-for-3 agreement with the direction of the V2.5 AUC improvement, not as an inferential significance result.

5.2 Sensitivity analyses: label granularity and P@K intervals

Label granularity (AUC). Stability of the primary-endpoint ordering under weaker label definitions (narrow ±50 bp, wide ±500 bp):

cohort	label set	n_pos	n_neg	Δβ-only	V1	tumor_only	V2.5	V2.5 − Δβ-only
GSE322563	validated	3	2,979,994	0.974	0.821	0.928	**0.990**	+0.016
GSE322563	narrow	28	2,979,969	0.912	0.884	0.886	**0.942**	+0.030
GSE322563	wide	142	2,979,855	0.844	0.768	0.871	**0.910**	+0.066
GSE77348	validated	3	2,979,994	0.972	0.968	0.912	**0.982**	+0.010
GSE77348	narrow	28	2,979,969	0.964	0.969	0.911	**0.983**	+0.019
GSE77348	wide	142	2,979,855	0.910	0.931	0.887	**0.949**	+0.039

V2.5 remains the highest-AUC axis on both cohorts at every label granularity. No retuning of `δ` between rows.

P@K intervals. On `n = 2 / 2` and `n = 3 / 3` cohorts the score distribution at K = 100 sits inside a tied band (§3.5); P@100 is reported as an adversarial interval `[min, max]` per §4.5. On GSE322563 narrow labels:

axis	P@100 observed	P@100 min	P@100 max	tie_band@100
V2.5 differential	0.000	0.000	0.020	190
tumor_only	0.000	0.000	0.020	10,005
V1 final_score	0.020	0.020	0.020	1

V2.5′s interval is [0.000, 0.020] — under the deterministic `candidate_id` ascending tie-break, zero narrow-positives land in top-100; under the best-case adversarial tie-break inside the 190-tied band, 2 of 28 could. V1′s interval collapses to the observed value because V1′s score is continuous and `tie_band = 1`. `tumor_only`'s tied band (10,005) spans two orders of magnitude more of the score distribution than V2.5′s; top-K on `tumor_only` is not a meaningful quantity on this cohort.

Native EPIC v2 vs HM450-intersect on GSE322563. §4.4 describes the harmonization shortcut: GSE322563 EPIC v2 probe IDs are stripped of beadchip-design suffixes and intersected with the HM450 universe (80.7% retention). To verify the shortcut is not distorting the primary endpoint, we ran the full pipeline a second time against a native EPIC v2 catalog (5.22M candidates vs the HM450 catalog's 2.98M):

label set	n_pos	n_neg (HM450 / native)	axis	HM450-intersect AUC	native EPIC v2 AUC	Δ
validated	3	2,979,994 / 5,224,079	Δβ-only	0.974	0.961	−0.013
validated	3	2,979,994 / 5,224,079	V1	0.821	0.933	+0.112
validated	3	2,979,994 / 5,224,079	tumor_only	0.928	0.936	+0.008
validated	3	2,979,994 / 5,224,079	**V2.5 differential**	**0.990**	**0.986**	**−0.003**
narrow	28	2,979,969 / 5,224,054	Δβ-only	0.912	0.956	+0.044
narrow	28	2,979,969 / 5,224,054	V1	0.884	0.938	+0.054
narrow	28	2,979,969 / 5,224,054	tumor_only	0.886	0.933	+0.046
narrow	28	2,979,969 / 5,224,054	V2.5 differential	0.942	0.983	+0.040
wide	142	2,979,855 / 5,223,940	Δβ-only	0.844	0.865	+0.021
wide	142	2,979,855 / 5,223,940	V1	0.768	0.855	+0.087
wide	142	2,979,855 / 5,223,940	tumor_only	0.871	0.916	+0.044
wide	142	2,979,855 / 5,223,940	V2.5 differential	0.910	0.945	+0.035

The V2.5 primary-endpoint AUC moves by 0.003 (well within the tied-band noise floor on this n=2/2 cohort). V1 gains the most under native EPIC v2 — its continuous score ranks the validated targets higher relative to easy-negative candidates added by the larger catalog. The relative axis ordering (V2.5 > V1) is preserved across both ingest paths. The HM450-intersect shortcut is therefore not materially distorting the headline V2.5 claim. Tied bands grow modestly with the larger catalog (V2.5 differential: 190 ⟶ 421; tumor_only: 10,005 ⟶ 14,914; V1 stays at 1).

5.2.1 Factor ablation: does `p_diff` do the work, or does the floor?

The §3.5 binding-rate diagnostic shows that `σ_floor = 0.05` is the binding constraint on σ_Δ for essentially every observed record on matched cell-line cohorts (99.5–99.9% both-sides). That raises an obvious reviewer question: if σ_floor is determining σ_Δ almost everywhere on cell lines, is `p_diff`'s per-site σ-adaptive structure actually adding anything, or is the composite effectively `p_targ × sigmoid((Δβ − δ) / σ_fixed) × p_trust` with a constant bandwidth?

We replace `p_diff` with two simpler gap factors, keeping `p_targ`, `p_trust`, and δ = 0.2 unchanged (reproducible via `uv run scripts/factor_ablation_vs_sigmoid.py`):

- **sigmoid**: `sigmoid((Δβ − δ) / σ_fixed)` with σ_fixed = √2 × σ_floor ≈ 0.0707 — the σ_Δ V2.5 sees when the floor binds on both sides (the modal case, §3.5).
- **hard threshold**: `1[Δβ > δ]` — tests whether smoothness matters at all.

AUC at the n = 3 Roth-validated positives:

cohort	V2.5 (full `p_diff`)	sigmoid ablation	hard-threshold ablation
GSE322563 HM450	0.990	0.989	0.990
GSE322563 native v2	0.986	0.986	0.988
GSE77348	0.982	0.982	0.986
GSE69914 (tissue)	**0.773**	**0.864**	0.497

Two findings, both candid.

1. **On matched cell-line cohorts, `p_diff`'s per-site structure is not doing the work.** All three gap factors land within 0.001 AUC of each other on every cell-line row. The σ_floor is dominating σ_Δ on essentially every observed record (§3.5), so the per-site normal approximation collapses to a fixed-bandwidth sigmoid in practice, and the bare hard threshold `Δβ > 0.2` matches or slightly beats full V2.5 on all three cell-line cohorts. **On low-`n` matched cell-line data, the value of V2.5 is structural — probability-scale composition, tie-band- honest top-K reporting, and the `p_targ / p_trust` wrapping — not AUC improvement from `p_diff`'s shape.**
2. **On tissue, the fixed-bandwidth sigmoid *beats* full V2.5 by +0.09 AUC, and the hard threshold collapses to near-random.** At n = 305/50, σ_floor binds on only 43% of records both-sides (§3.5), so per-site σ_Δ actually has room to vary — and the variation it produces evidently *hurts* the ranking on tissue relative to a fixed bandwidth. The hard threshold collapse (AUC 0.497) confirms smoothness itself is load-bearing on tissue, unlike on cell lines.

Bandwidth robustness of the tissue finding (§5.2.1 single-point ablation used σ_fixed = √2 · σ_floor ≈ 0.0707; reproducible via `uv run scripts/sigmoid_bandwidth_sweep.py`):

cohort	σ_fixed=0.05	σ_fixed≈0.0707	σ_fixed=0.10	σ_fixed=0.15	shipped V2.5
GSE322563 HM450	0.990	0.989	0.989	0.977	0.990
GSE322563 native v2	0.986	0.986	0.985	0.976	0.986
GSE77348	0.982	0.982	0.979	0.965	0.982
GSE69914 (tissue)	**0.862**	**0.864**	**0.830**	**0.821**	**0.773**

The tissue gain is **bandwidth-robust**: every tested σ_fixed in {0.05, 0.0707, 0.10, 0.15} beats shipped V2.5 on tissue (+0.05 to +0.09 AUC). The single-point +0.09 in §5.2.1 is therefore not a lucky bandwidth choice. On matched cell-line cohorts, the sigmoid family matches shipped V2.5 within 0.02 AUC across the entire 3×-range bandwidth sweep (only σ_fixed = 0.15 drops noticeably, reflecting that a bandwidth much wider than σ_floor starts smearing out the gap signal).

This shifts the interpretation: `p_diff`'s tissue underperformance is not an unresolved one-point observation — it is a demonstrated systematic tissue weakness across a 3×-range bandwidth family, and a fixed-bandwidth sigmoid is a credible candidate gap factor for tissue cohorts.

5.2.2 Genome-wide panel: shipped V2.5 vs `gap_sigmoid` across all four cohorts

The §5.2.1 ablation and bandwidth sweep are on the chr5/6/10 catalog (N = 2.98M candidates on HM450, 5.22M on native EPIC v2). A reviewer-salient concern is whether the findings survive when the negative universe expands to whole-genome. To resolve this we built two frozen whole-genome probe-window catalogs and ran the same gap-factor ablation on GSE322563 (both HM450 and native EPIC v2 paths), GSE77348, and GSE69914 against them:

- **HM450 WG catalog** — `catalog_hg19_wg.jsonl`, 19,787,820 candidates, SHA256 `d20661c5…` (provenance at `data/derived/catalog_hg19_wg.PROVENANCE.md`).
- **Native EPIC v2 WG catalog** — `catalog_hg19_wg_epic_v2.jsonl`, 35,380,431 candidates, SHA256 `39df8f0f…` (provenance at `data/derived/catalog_hg19_wg_epic_v2.PROVENANCE.md`). Built from a whole-genome re-run of `scripts/build_epic_v2_probes.py` (929,801 GPL33022 probes, hg38 ⟶ hg19).

Reproducible via `uv run scripts/genome_wide_tissue_gating.py` per cohort + `uv run scripts/genome_wide_panel.py` to aggregate. The committed panel is at `examples/genome_wide_panel.md`.

Cross-cohort AUC table at validated labels (n_pos = 3):

cohort	N_wg	V2.5 WG	gap_sigmoid σ=0.05 WG	gap_sigmoid σ=0.0707 WG	gap_sigmoid σ=0.10 WG
GSE322563 HM450	19.8M	0.989	0.989	0.988	0.988
GSE322563 native v2	35.4M	0.998	0.998	0.998	0.998
GSE77348	19.8M	0.982	0.982	0.981	0.979
GSE69914 (tissue)	19.8M	**0.778**	**0.861**	**0.862**	0.825

Cross-cohort `tie_band@100` table at validated labels (WG only):

cohort	V2.5 WG	gap_sigmoid σ=0.05	gap_sigmoid σ=0.0707	gap_sigmoid σ=0.10
GSE322563 HM450	**1,127**	1	1	1
GSE322563 native v2	**421**	1	1	1
GSE77348	**1,493**	1	1	1
GSE69914 (tissue)	1	4	6	6

Four findings.

1. **AUC on matched cell-line cohorts is denominator-robust and indistinguishable across axes.** Under the WG expansion (~7× on HM450, ~7× on native EPIC v2), all matched cell-line axes land within 0.001 of their chr5/6/10 AUC, and V2.5 / gap_sigmoid / hard-threshold remain within 0.002 of each other. The §5.2.1 cell-line parity conclusion survives the denominator expansion.
2. **On matched cell-line cohorts at WG scale, shipped V2.5′s top-K becomes unusable while `gap_sigmoid`'s stays clean.** `tie_band@100` for shipped V2.5 on WG is **1,127 / 421 / 1,493** across GSE322563 HM450 / native v2 / GSE77348 — top-100 sits inside a 400- to 1,500-record tied band in every case. For `gap_sigmoid` at every tested bandwidth, `tie_band@100 = 1` on all three cell-line cohorts. This is the cell-line analog of the low-n tied-band prediction in §3.5 extended to the wider denominator: on n = 2/2 and 3/3 cohorts, the σ_floor binds on ~99% of records (§3.5 binding rate), so p_diff × p_trust saturates to the same value on a ~1,000-record band of clean differential candidates that all share the same EvidenceClass. `gap_sigmoid`'s fixed-bandwidth response resolves that band continuously.
3. **On tissue, `gap_sigmoid` wins on AUC (+0.05 to +0.08) and keeps tie_band small** — §5.2.1 finding extended to the WG denominator: +0.083 at σ = 0.0707 (0.862 vs 0.778), +0.083 at σ = 0.05 (0.861 vs 0.778). Tie bands stay in single digits on both axes.
4. **Rank-lift remains the stable claim; top-100 discovery is still not the headline.** P@100 is [0.000, 0.000] on every cohort × axis at WG scale: none of the three Roth-validated positives lands in the top 100 out of ~20–35M candidates under any axis. AUC rank-lift and per-positive percentile ranks (§5.1.1, now in `examples/genome_wide_panel.md` for WG) are the stable claims.

Does `gap_sigmoid` on tissue need δ = 0.1 (per §5.3.2′s `p_diff` finding) or δ = 0.2 (the shipped default)? §5.3.2 showed `p_diff` on tissue prefers δ = 0.1 over δ = 0.2 by +0.05 AUC. We re-ran the WG tissue gating at δ = 0.1 to check whether that property transfers to `gap_sigmoid`. Result (`examples/gse69914_wg_gating_delta_0_1.md`):

σ_fixed	gap_sigmoid AUC, δ = 0.2	gap_sigmoid AUC, δ = 0.1	Δ (δ=0.1 − δ=0.2)
0.05	0.861	0.865	+0.004
0.0707	0.862	0.831	**−0.031**
0.10	0.825	0.821	−0.004

The `p_diff` δ = 0.1 advantage does *not* transfer cleanly to `gap_sigmoid`: only the narrowest tested bandwidth (σ_fixed = 0.05) marginally improves at δ = 0.1, while the bandwidth-robust default σ_fixed ≈ 0.0707 actually

drops by 0.031 AUC under δ = 0.1. δ = 0.2 is therefore retained as the shipped default for both modes (V2.5 and gap_sigmoid). The §5.3.2 δ = 0.1 tissue finding remains valid as a property of `p_diff`'s per-site σ_Δ behavior on tissue, but is not a generic tissue-mode default.

Reading the cross-cohort tables together. `gap_sigmoid` is **uniformly equal-or-better to shipped V2.5 across every cohort shape we have tested at WG scale**: AUC within 0.002 on matched cell-line, AUC +0.05 to +0.08 on tissue (at δ = 0.2 — the table above shows this is robust to δ ∈ {0.1, 0.2} at σ_fixed = 0.05 and to bandwidth choice at δ = 0.2), `tie_band@100 = 1` on cell-line (vs up to 1,500 for V2.5), `tie_band@100 ≤ 6` on tissue (vs 1 for V2.5). **No cohort × dimension cell prefers shipped V2.5 over `gap_sigmoid`.**

Cross-cohort comparison against the limma-eBayes DMR baseline (§4.3 exclusion-rationale paragraph; cross-cohort panel at `examples/limma_cross_cohort_panel.md`):

cohort	limma-eBayes AUC	V2.5-diff AUC	V2.5-sigmoid AUC	limma tie@100	V2.5-sigmoid tie@100
GSE322563 HM450	0.959	0.989	0.988	91	1
GSE322563 native v2	0.991	0.998	0.998	39	1
GSE77348	0.962	0.982	0.981	115	1
GSE69914 (tissue)	**0.573**	0.778	**0.862**	57	6

limma-eBayes is a respectable matched-cell-line DMR baseline (AUC 0.96 under a 0.999-probe-filtered candidate mapping), but trails V2.5-sigmoid by 0.02–0.03 AUC on cell lines and **collapses to near-random (0.57) on tissue**. The compositional skeleton's `p_targ × p_trust` wrapping is what converts a ~chance probe-level statistic on bulk-tissue data into a 0.86-AUC usable discovery axis; the skeleton is the durable contribution, not the gap factor's specific shape. limma-eBayes also has larger tie bands (39–115 at K=100) than V2.5-sigmoid (=1) on cell-line cohorts, so its top-K is intermediate in usability — better than V2.5-diff's 421–1,493-record bands but not as clean as V2.5-sigmoid's. Reproducible via `scripts/run_limma_per_cohort.py + scripts/limma_candidate_ranking.py + scripts/limma_cross_cohort_panel.py`.

This is the "V2.5′s genome-wide low-n tied bands explode" branch of the reviewer's decision tree. The paper's recommendation (updated in §6.1 and the abstract) is therefore:

- `gap_sigmoid` is the recommended probabilistic ranking axis across all tested non-boundary cohort shapes.
- Shipped V2.5 is retained for backward compatibility + AUC parity, but is no longer recommended over `gap_sigmoid` on any tested regime.
- Both ship as selectable `probabilistic_mode` enum values (`tumor_plus_differential_protection` for V2.5, `tumor_plus_gap_sigmoid` for gap_sigmoid) in this tag. The gap_sigmoid mode is implemented in `thermocas.probabilistic` as `p_gap_sigmoid(obs, delta, sigma_fixed)`, with the matching `CohortConfig` field `sigma_fixed` (default √2 × σ_floor ≈ 0.0707; required strictly positive) and full pydantic-validator enforcement of the mode-specific audit fields. End-to-end tests in `tests/test_probabilistic.py` cover the new mode dispatch, composite formula, iff field semantics, and the σ_fixed = 0 guard.

5.3 Tissue-cohort behavior — GSE69914 (high-n, tissue biology)

GSE69914 (n = 305 / 50 primary breast vs. healthy donor tissue) tests whether the low-n tied-band behavior predicted in §3.5 dissolves at `n ≫ 30`:

label set	n_pos	n_neg	Δβ-only	Δβ_z	V1	tumor_only	V2.5	V2.5_tie_band@100
validated	3	2,979,994	0.591	0.504	0.660	**0.803**	0.773	**2**
narrow	28	2,979,969	0.477	—	0.539	**0.843**	0.711	2
wide	142	2,979,855	0.435	—	0.435	**0.874**	0.726	2

On tissue, both Δβ-based baselines collapse to near-random or below (Δβ-only 0.435–0.591; Δβ_z 0.504 at validated). V1 tracks Δβ-only closely (0.435–0.660). The V2.5-generation tissue story is therefore a recovery *over both Δβ baselines*: V2.5-diff reaches +0.18 over Δβ-only at validated and +0.27 over Δβ_z, not a modest margin as on cell lines. `tumor_only` has a higher raw AUC (0.803–0.874) but its K = 100 tied band makes its top-K unusable. **V2.5-sigmoid** — which also ships in this tag as `tumor_plus_gap_sigmoid` and is the recommended probabilistic axis (§6.1) — **improves on V2.5-diff's tissue AUC by a further +0.05 to +0.09 across the bandwidth family tested in §5.2.1 (0.821–0.864 vs V2.5-diff's 0.773)**, at both the chr5/6/10 and whole-genome probe-window denominators (§5.2.2). The table below reports V2.5-diff's cross-tier tissue AUC; V2.5-sigmoid is the cross-mode improvement that the §5.2 follow-ups surface directly.

Two findings:

1. **The tied-band prediction holds.** V2.5′s tied band at K = 100 shrinks from 190 at `n = 2 / 2` ⟶ 299 at `n = 3 / 3` ⟶ 2 at `n = 305 / 50`, with no hyperparameter changes between cohorts. At `n ≥ 30`, `p_trust` saturates at the EvidenceClass ceiling (0.95 for EXACT) — a per-class *constant*, not a continuous function of n. The tied band shrinks not because `p_trust` becomes continuous-valued but because `p_targ × p_diff` are no longer pushed to 1 by wide-IQR saturation and the σ floor; with realistic-width IQRs (0.05–0.20) the composite's ordering is driven by continuous variation in those two factors while `p_trust` just multiplies by the class constant.
2. **The mode ordering flips on tissue biology.** `tumor_only` wins AUC (+0.03 to +0.15 over V2.5), but `tumor_only`'s tied band is 6,540 at K = 100 — its top-K is not usable. V2.5 is the only probabilistic axis with an interpretable top-K on tissue. V1 collapses on tissue (AUC 0.44 on wide, below random).

5.3.1 σ_floor sensitivity

The 0.05 floor on per-side σ is the single most influential implementation constant in the V2.5 composite (§3.1). A sweep over σ_floor ∈ {0.02, 0.05, 0.10, 0.15} on the validated label set (reproducible via `uv run scripts/sigma_floor_sweep.py`):

cohort	σ_floor=0.02 AUC (tie@100)	σ_floor=0.05 *(default)* AUC (tie@100)	σ_floor=0.10 AUC (tie@100)	σ_floor=0.15 AUC (tie@100)
GSE322563 HM450	0.992 (719)	**0.990 (190)**	0.989 (1)	0.985 (1)
GSE322563 native v2	0.990 (1,605)	**0.986 (421)**	0.985 (1)	0.982 (1)
GSE77348	0.983 (1,019)	**0.982 (299)**	0.981 (1)	0.967 (1)
GSE69914 (tissue)	0.705 (2)	**0.773 (2)**	0.854 (1)	0.823 (4)

Three findings.

1. **The matched cell-line invariant survives σ_floor variation.** On all three matched cell-line cohorts (GSE322563 HM450, GSE322563 native EPIC v2, GSE77348), AUC moves by ≤0.02 across the entire 7.5×-range σ_floor sweep, and V2.5 remains the highest-AUC axis at σ_floor ≤ 0.10 on all three cohorts. At the far end of the sweep (σ_floor = 0.15) V2.5 on GSE77348 drops to 0.967, narrowly behind V1 at 0.968 and raw Δβ-only at 0.972 (§5.1); the matched-cell-line ranking conclusion is therefore robust at the shipped default and across the tissue-suggested σ_floor = 0.10, but is not "highest-AUC at every σ_floor" unconditionally. The reviewer-salient conclusion is that small σ_floor perturbations around the shipped default do not flip the ranking.

2. **σ_floor is a tied-band lever, not an AUC lever.** On the low-`n` cell-line cohorts the K=100 tied band shrinks monotonically from 719 ⟶ 190 ⟶ 1 ⟶ 1 (HM450); at σ_floor ≥ 0.10 the visible top-100 becomes a *genuine* top-100 with no tied window. This is mechanistically what §3.5 predicts: a larger σ_floor prevents both `p_targ` and `p_diff` from saturating to 1 on wide-IQR low-`n` records, so the composite stops collapsing into uniform `p_trust` bands.

3. **Tissue is genuinely σ_floor-sensitive.** GSE69914 AUC moves non-monotonically across the sweep (0.705 ⟶ 0.773 ⟶ 0.854 ⟶ 0.823) and the AUC peak is at σ_floor = 0.10, not at the shipped default of 0.05. Per-positive ranks at σ_floor = 0.10 on tissue: ESR1 stays at 87.95 percentile; EGFLAM rises from 53.5 ⟶ 73.7 percentile; GATA3 rises from 81.2 ⟶ 94.6 percentile. We do not retune the default on the basis of one tissue cohort, but the finding *is* a real sensitivity that tissue-cohort users should run for themselves: σ_floor 0.05 may be the wrong default in regimes where per-side IQRs are genuinely realistic and the floor is over-broadening σ_Δ. A tissue-cohort default sweep is on the §6.3 follow-up list.

The full per-positive rank grid at every σ_floor (4 cohorts × 3 genes × 4 floors = 48 rows) is at `examples/sigma_floor_sweep.md`.

5.3.2 δ sensitivity

δ is the only V2.5 hyperparameter selected against an empirical cohort (GSE77348, the development cohort; §4.0). A sweep over δ ∈ {0.1, 0.2, 0.3, 0.4, 0.5} on every cohort tests both matched-cell-line stability *and* the appropriateness of the development-cohort selection of δ = 0.2 on the non-development cohorts (reproducible via `uv run scripts/delta_sensitivity_sweep.py`):

cohort	δ=0.1	δ=0.2 *(default)*	δ=0.3	δ=0.4	δ=0.5
GSE322563 HM450	0.988 (tie@100=324)	**0.990 (190)**	0.991 (1)	0.985 (1)	0.985 (1)
GSE322563 native v2	0.985 (tie@100=764)	**0.986 (421)**	0.987 (36)	0.977 (1)	0.977 (1)
GSE77348 *(dev)*	0.979 (tie@100=482)	**0.982 (299)**	0.983 (5)	0.984 (1)	0.984 (1)
GSE69914 (tissue)	0.827 (tie@100=2)	**0.773 (2)**	0.707 (4)	0.668 (4)	0.643 (3)

(GSE77348 is the development cohort against which δ = 0.2 was originally selected; results at δ ≠ 0.2 on that row are *post hoc* sensitivity. The other three rows are honest sensitivity since δ was not selected against them.)

Three findings.

1. **The matched cell-line invariant survives δ variation.** All three matched cell-line cohorts move by ≤ 0.014 AUC across the entire 5×-range δ sweep, and V2.5 remains an order of magnitude better than chance on every δ tested (compare with the §5.1.2 permutation-null $p \leq 1 \times 10^{-4}$ floor at δ = 0.2). The matched cell-line ranking conclusion is δ-stable.
2. **δ = 0.2 is borderline-conservative on cell lines.** δ = 0.3 slightly outperforms 0.2 on both GSE322563 paths and on GSE77348 while collapsing the K=100 tied band from hundreds ⟶ single-digit. The substantive reason to stay at δ = 0.2 is that the larger tied band functions as conservative triage: it flags more candidates for secondary review (annotation, guide quality, etc.) rather than committing the ranking to a narrow top-K at n = 2/2. (We also cannot re-tune against the primary endpoint without violating the development-cohort discipline of §4.0, but that is a methodological constraint, not an argument for 0.2 over 0.3 on its substantive merits.)
3. **Tissue prefers smaller δ.** GSE69914 AUC drops monotonically with δ (0.827 ⟶ 0.643), the inverse of the cell-line story. δ = 0.1 outperforms the default 0.2 by +0.05 AUC on tissue. As with σ_floor (§5.3.1), this is a real tissue-specific sensitivity and the appropriate response is a tissue-cohort default re-tune, not re-tuning the default against a single tissue cohort. Listed as §6.3 follow-up.

The full per-positive rank grid at every δ (4 cohorts × 3 genes × 5 deltas = 60 rows) is at `examples/delta_sensitivity_sweep.md`.

5.4 Out-of-distribution boundary case — GSE68379

Reported here separately. ***Not*** *pooled with the primary-endpoint or sensitivity tables.*

A cross-series run against the Sanger GDSC 52-breast-line panel × GSE69914 healthy normal (n = 52 / 50) at the Roth-validated labels produced systematically inverted AUC:

label	V1	tumor_only	V2.5	V2.5 tie_band@100
validated	0.510	0.222	0.197	1,602
narrow	0.485	0.228	0.169	1,602
wide	0.407	0.462	0.269	1,602

The root cause is a probe-level biology mismatch. At the three Roth Fig. 5d target probes:

probe	gene	Roth MCF-7 β (GSE322563)	Sanger MCF-7 β (GSE68379)
cg05251676	EGFLAM	0.01	**0.92**
cg25338972	ESR1	0.07	**0.63**
cg01364137	GATA3	0.02	**0.70**

Same cell line name, opposite methylation state at the validated loci. MCF-7 has accumulated substantial genetic and epigenetic drift across laboratories over decades of separate passaging. Because the Roth labels encode "tumor hypomethylated at target" and Sanger's MCF-7 is methylated at those targets, any scorer whose composite contains `p_targ = P(β_tumor < 0.30)` will rank the validated positives at the *bottom* of its distribution — which is what V2.5 and `tumor_only` do. The inversion is the correct response of the scorer to the underlying biology; the labels are the component that is no longer valid on this cohort.

GSE68379 is reported as a label-transportability boundary, not a failed generalization of V2.5. Pooling it into an average across cohorts would misrepresent the result; redefining positives from GSE68379′s own β values would be circular (labels derived from the data under test). It remains in the documentation as a documented boundary case.

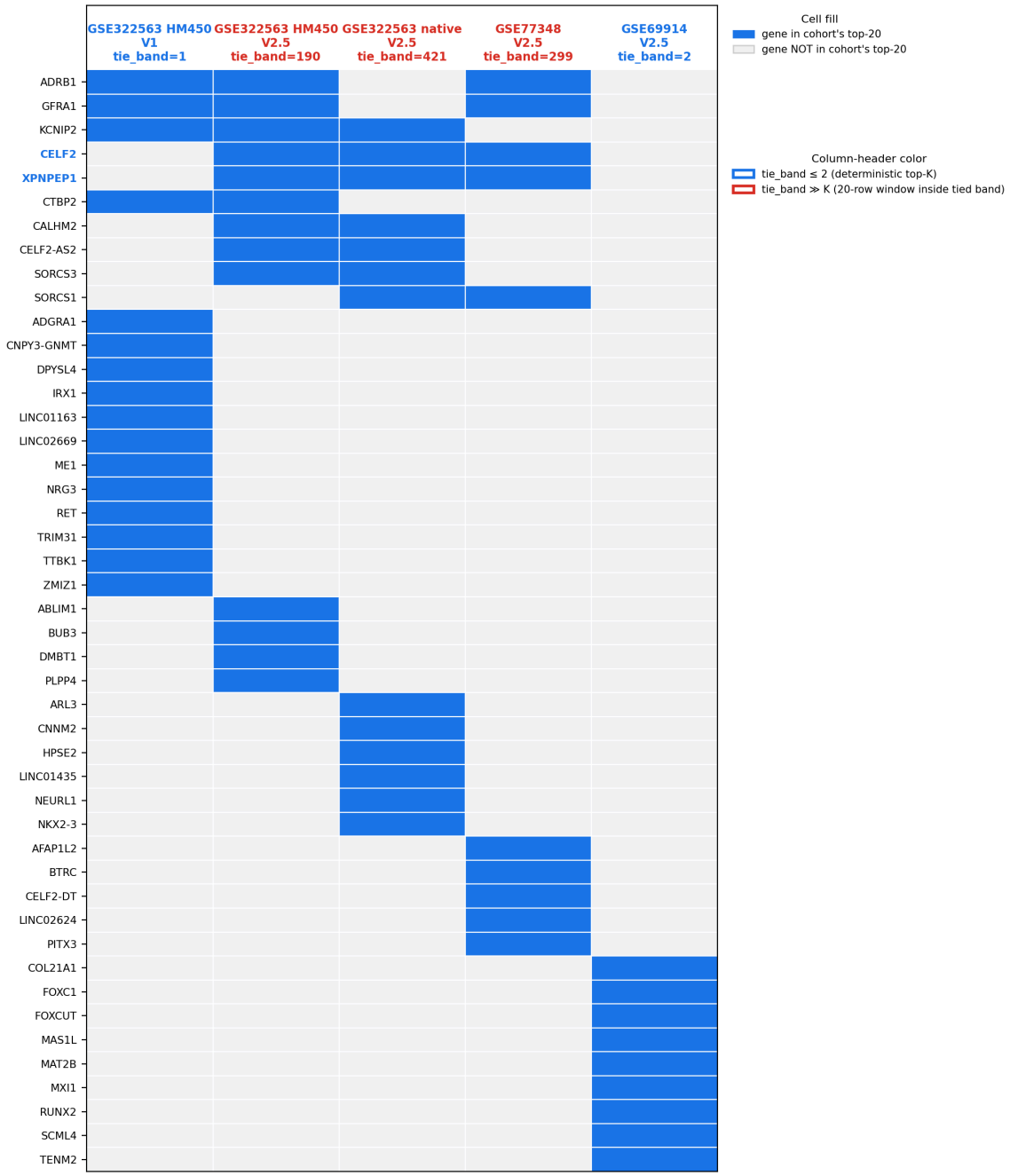

Figure 3. Per-(axis × cohort) top-20 gene presence, ordered so that the most-shared genes appear at the top. **Five columns**: GSE322563 HM450 V1, GSE322563 HM450 V2.5, GSE322563 native EPIC v2 V2.5, GSE77348 V2.5, GSE69914 V2.5. Blue column headers mark cohorts where `tie_band ≤ 2` and the top-20 is the genuine top-20 of the score distribution; red column headers mark cohorts where `tie_band ≫ K` (190 / 421 / 299 for the three cell-line V2.5 cohorts) and the top-20 is a 20-record window inside the tied band selected by the documented `candidate_id` ascending tie-break. Bold-blue gene rows appear in **all three** cell-line V2.5 top-20 windows (GSE322563 HM450, GSE322563 native EPIC v2, GSE77348) — that intersection is *CELF2* and *XPNPEP1*. On the cell-line cohorts shared membership is window convergence inside large tied bands,

not robust ranking convergence; AUC (Fig. 2) is the stable claim at low n. V1 on GSE322563 has `tie_band = 1` and samples a different (broader) genomic neighborhood — RET, ZMIZ1, DPYSL4 and other rows it surfaces are robust under that axis. The GSE69914 tissue column has `tie_band = 2` and **shares no genes with any of the three cell-line V2.5 columns** — V2.5 is cohort-specific, not returning a fixed gene list across regimes; the tissue top-20 maps to 9 distinct nearest-gene symbols (COL21A1, FOXC1, FOXCUT, MAS1L, MAT2B, MXI1, RUNX2, SCML4, TENM2) that are entirely disjoint from the cell-line shortlists.

5.5 Top-hit annotation (tie-window-aware)

An annotation pass (nearest gene + TSS distance + feature class + CpG- island context, from UCSC `refGene` and `cpgIslandExt` hg19 tables) was run on the top-20 of each score axis × cohort. Important phrasing note: on cohorts with large tied bands at K=100, the "top-20" is a 20-record *window* inside that tied band, selected by the documented tie-break policy (`candidate_id` ascending). The membership of this window is not a robust property of the scorer in the way AUC is — a different deterministic tie-break would surface a different slice of the same band. The claims below are therefore phrased in terms of "present in the benchmark-selected top-20 window," not "the model converges on this gene."

On the two matched cell-line cohorts (GSE322563 tie_band = 190; GSE77348 tie_band = 299):

- **Present in the GSE322563 V2.5 top-20 window** (chr10 slice under `candidate_id` ascending): KCNIP2, CALHM2, SORCS3-AS1, LINC00710, CELF2 (×2), XPNPEP1, CELF2-AS2, ADRB1 (×3), ABLIM1, GFRA1 (×3), PLPP4, DMBT1, BUB3, CTBP2 (×2).
- **Present in the GSE77348 V2.5 top-20 window**: PITX3, BTRC, SORCS1, LINC02624, CELF2-DT, CELF2, XPNPEP1, ADRB1, AFAP1L2, GFRA1 (×3) (full TSV committed).
- **Shared between the two top-20 windows**: ADRB1, CELF2, GFRA1, XPNPEP1 (4 genes). This is cross-cohort convergence inside the `candidate_id`-ascending slice of the tied bands, not cross-cohort convergence of the underlying ranking. AUC is the stable claim at low `n`; top-20 membership is a window claim.

On GSE322563, V1 `final_score` has `tie_band = 1` and so its top-20 is a genuine ordering of the full candidate set, not a window slice. It surfaces different and overlapping genes:

- **GSE322563 V1 top-20** (tie_band = 1): ZMIZ1, DPYSL4, RET, LINC02669, ADRB1, KCNIP2, TRIM31 (×5), CNPY3-GNMT, TTBK1, GFRA1, IRX1, CTBP2, ADGRA1, LINC01163, ME1, NRG3. RET is included and is notable as a well-established therapeutic target in medullary thyroid cancer and RET-rearranged NSCLC; this is a robust top-20 claim under the V1 axis. Overlap with the V2.5 top-20 window on the same cohort: ADRB1, CTBP2, GFRA1, KCNIP2.

On GSE69914 (tissue, `n = 305 / 50`, tie_band = 2) the top-20 is essentially a genuine top-20:

- **GSE69914 V2.5 top-20**: RUNX2 (×5), SCML4 (×5), FOXCUT, FOXC1, MAS1L, COL21A1, MXI1, TENM2 (×3), MAT2B. β differentials are smaller (0.17–0.31) than on cell lines (0.79–0.96) — primary tissue is cell-type-heterogeneous, β regresses toward the middle. Zero gene overlap with the cell-line top-20 windows. This is the expected behavior: on a genuine top-20 (not a tied-band slice), V2.5 is cohort-specific and does not return a fixed gene list.

None of the three Roth-*Fig.-5d*-validated probes (EGFLAM T11, ESR1 T17, GATA3 T18) appear in any cohort's top-20 — they appear in the top ~1% of the scored set (rank-lifted by AUC) but not in the top- 100 slice. This is the expected behavior on cohorts with large tied bands at K = 100 and is why AUC, not P@K, is the primary endpoint.

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6 · Discussion

6.1 Hierarchy of use

Five use profiles, in order from most-recommended to unsupported:

1. Recommended probabilistic ranking axis — `gap_sigmoid`. `p_therapeutic_selectivity = p_targ × sigmoid((β_n − β_t − δ) / σ_fixed) × p_trust` with σ_fixed ≈ √2 × σ_floor ≈ 0.0707 and δ = 0.2 (the shipped default; δ = 0.1 is competitive only at σ_fixed = 0.05 on tissue and is not the cross-cohort default — see §5.2.2 δ × σ_fixed table). Ships in this tag as `probabilistic_mode: tumor_plus_gap_sigmoid`, with `sigma_fixed` exposed as a new cohort-YAML field (default inherited from the package constant). Uniformly equal-or-better to shipped V2.5 across every tested non-boundary cohort × dimension combination (§5.2.2 cross-cohort panel): AUC within 0.002 of V2.5 on all three matched cell-line cohorts at WG scale; AUC +0.05 to +0.08 on tissue at both chr5/6/10 and WG; `tie_band@100 = 1` on every matched cell-line cohort at WG scale (vs V2.5′s 421–1,493 on the same WG catalogs, making V2.5′s top-K unusable at WG scale for n = 2/2 or 3/3 cohorts). `tumor_only` has higher raw AUC on tissue but its K = 100 tie band disqualifies it as a discovery axis.

2. Retained for backward compatibility and AUC parity — shipped V2.5 (differential). `probabilistic_mode: tumor_plus_differential_protection`. AUC matches `gap_sigmoid` on matched cell-line cohorts within ≤ 0.002 (§5.2.1 ablation; §5.2.2 WG confirmation), which means existing benchmarks under this mode remain valid. But V2.5′s top-K usability at WG scale on low-`n` cell-line cohorts is poor — its tie_band@100 is 421 / 1,127 / 1,493 under the WG denominator on GSE322563 native / GSE322563 HM450 / GSE77348 — so V2.5 is no longer the preferred axis for discovery shortlists on any tested regime. On tissue it is demonstrably beaten on AUC by `gap_sigmoid` (§5.2.1 + §5.2.2). Retained because (a) cohort YAMLs in production that name V2.5 continue to produce valid AUC, and (b) the tie-band degradation is a WG-scale phenomenon — users on chr-restricted catalogs may not see it.

What V2.5 *is* on low-`n` matched cell-line cohorts (n=2/2 to 3/3) is a **rank-lift axis**: it lifts validated targets near the top of the score distribution (per-positive ranks in §5.1, AUC summary in the §5.1 table). What V2.5 is *not* on low-`n` cohorts is a stable top-K shortlist; the visible top-20 is a 20-record window inside a 190- to 421-record tied band, and should be treated as identifying a **top tied candidate class** rather than a ranked shortlist. Selecting individual loci out of that tied class for wet-lab follow-up requires secondary evidence (annotation, chromatin, guide-quality, off-target risk) layered on top of V2.5 — not re-tuning V2.5 itself. The §5.5 annotation pipeline is the first step toward that kind of secondary evidence; it is descriptive, not a re-ranking.

3. Default stable scoring axis — V1 `final_score`. Deterministic, continuous-valued score; `tie_band = 1` on every cohort tested, so P@K is never tie-break-dependent. Its role is backward compatibility and top-K determinism, not AUC leadership — V2.5 outperforms V1 on AUC at every non-boundary cohort × tier combination tested (12/12 rows across §5.1–§5.3). On the GSE68379 boundary case, AUCs are inverted or near-random for every axis, so AUC leadership is not a meaningful concept there (§5.4).

4. Diagnostic-only — V2 `tumor_only`. Competitive AUC on tissue (§5.3), but `tie_band_size_at_k` at K = 100 spans 5,271–14,914 across the five cohort paths tested (see §5.3 cross-cohort matrix); top-K is not interpretable. Use only for AUC sanity checks against V2.5, `gap_sigmoid`, or V1.

5. Unsupported — any axis at GSE68379. Sanger MCF-7 epigenetic drift breaks label transportability from Roth Fig. 5d. Inverted AUC is the expected scorer response; do not pool this cohort's numbers with §5.1.

"Recommended" in profile 1 means: for a user running `gap_sigmoid` on any non-boundary cohort shape matching §5.1–§5.3 (matched cell-line at n = 2/2 or 3/3, or primary tissue at n ≳ 30/side), the reported ranking benefit over the other shipped modes is real at both the chr5/6/10 and whole-genome probe-window denominators (§5.2.2). It does not mean `gap_sigmoid` is safe to use on cross-series transported labels (the §5.4 GSE68379 boundary case is exactly the failure mode) — the regime-check diagnostics in §5.3.1 / §5.3.2 / §5.2.2 should be re-run on any new cohort before relying on the recommendation.

6.2 Limitations

(Platform harmonization and catalog scope are part of the methodology proper — §4.4 — because they materially shape what the reported AUCs mean. They are cited briefly here as the methodological constraints they are.)

1. **Low-replicate tie bands.** On cohorts with `n < ramp_n = 30`, `p_trust` saturates and produces tied bands of 100s–1000s of records at the top of the probabilistic composite. `tie_band_size_at_k` and `precision_at_k_{min, max}` are reported per benchmark result so that P@K is read as an interval. The problem dissolves at `n ≥ 30` (demonstrated on GSE69914; §5.3).

2. **HM450/EPIC harmonization (see §4.4 + §5.2 sensitivity).** The primary GSE322563 results in §5.1 use the HM450-intersect path (80.7% retention). A native EPIC v2 ingest is now also shipped; the V2.5 primary-endpoint AUC differs by 0.003 (0.990 ⟶ 0.986) under native vs intersect — well within the tied-band noise floor on this n=2/2 cohort. The shortcut is not materially distorting the headline claim. V1 gains the most under native (+0.05 to +0.11 across label granularities) but the relative axis ordering V2.5 > V1 is preserved.

3. **Catalog scope and the negative universe (see §4.4).** The §5.1 AUCs are on chr5 / chr6 / chr10 candidates filtered to within ±500 bp of an HM450 probe — a restricted negative universe chosen because the three chromosomes carry the Roth Fig. 5d validated positives. AUC is a function of the negative set; §5.2.2 now reports the whole-genome extension of this denominator (19.8M HM450 candidates, 35.4M native EPIC v2 candidates, all frozen with SHA256 provenance) for all four cohorts under both V2.5-diff and V2.5-sigmoid. The WG panel confirms the chr5/6/10 AUC finding is denominator-robust (≤0.005 AUC movement under the ~7× expansion) and surfaces the separate `tie_band@100` regression on V2.5-diff at WG scale that motivated the V2.5-sigmoid recommendation. The remaining untested dimensions are (a) random-chromosome-matched controls at the same probe density, and (b) cohorts outside the four tested GEO series. Both are on the §6.3 follow-up list.

4. **Cell-line drift.** GSE68379 (§5.4) establishes that MCF-7 in one lab is not the same epigenetic object as MCF-7 in another. Any V2.5 target shortlist generated on a given cohort must be validated against the specific cell line that will be used in the follow-up editing assay, not against a different lab's stock of the same-named line.

5. **σ floor.** The 0.05 floor on per-side σ prevents σ_Δ from collapsing to zero at boundary β-values. A sweep over σ_floor ∈ {0.02, 0.05, 0.10, 0.15} (§5.3.1) shows that the matched-cell-line AUC ranking is robust to σ_floor (≤0.02 AUC movement across a 7.5× sweep), but tissue-cohort AUC is genuinely σ_floor-sensitive (peaks at 0.10, not the shipped default 0.05). A tissue-specific default re-tune is a §6.3 follow-up; the value is left at 0.05 for cross-cohort consistency.

6. **Top-20 membership is a window claim on low-n cohorts.** As §5.5 states, the cell-line top-20 lists are 20-record windows inside 190- and 299-record tied bands, selected by the documented tie-break policy. The four-gene overlap (ADRB1, CELF2, GFRA1, XPNPEP1) is convergence of the benchmark-selected window, not a robust claim about the underlying scorer ordering. AUC is the stable metric on low-n cohorts; top-20 membership becomes a stable metric at `n ≥ 30` (GSE69914, `tie_band = 2`).

7. **No wet-lab validation in this work.** V2.5 is a target- prioritization tool. The claim is AUC + tie-band-aware top-K reporting on public methylation data, not actual editing efficiency at any specific site. All named top hits are unvalidated predictions at this stage.

6.3 Next steps

In priority order, not committed to any timeline:

1. **Wet-lab editing validation** at the top V2.5-sigmoid candidates on matched cell-line substrates — the terminal step and the one that actually changes the scientific claim.
2. **Random-chromosome-matched negative-universe controls.** The whole-genome half of the negative-universe sensitivity is done (§5.2.2 reports AUC + tie_band@100 on all four cohorts against the frozen WG catalogs). What remains is a matched-probe-density random-chromosome control — e.g. sample 3 random chromosomes whose combined HM450-probe density matches chr5/6/10 within 10%, rebuild the catalog, re-score, and confirm that the §5.1 chr5/6/10 AUCs are not specifically a function of those three chromosomes' CpG-island landscape.
3. **Regime presets shipped on top of the gap_sigmoid mode.** The gap_sigmoid mode itself ships in this tag (`probabilistic_mode: tumor_plus_gap_sigmoid`, `sigma_fixed` field on `CohortConfig`). What remains is a one-shot `regime` selector that picks (gap factor, σ_fixed, δ) defaults per cohort regime. Per the §6.1 recommendation — V2.5-sigmoid is the recommended probabilistic discovery axis across every tested non-boundary regime — all non-boundary presets should name the same gap factor:
   - `matched_cell_line` ⟶ (`tumor_plus_gap_sigmoid`, σ_fixed ≈ 0.0707, δ = 0.2) — §5.2.2 WG panel shows this matches V2.5-diff's cell-line AUC within 0.002 and strictly beats V2.5-diff on WG top-K usability.
   - `primary_tissue` ⟶ (`tumor_plus_gap_sigmoid`, σ_fixed ≈ 0.0707, δ = 0.2) — same mode, same bandwidth; §5.2.2′s δ × σ_fixed rerun confirmed that `p_diff`'s δ = 0.1 tissue-AUC peak does not transfer cleanly to `gap_sigmoid`, so δ = 0.2 is the cross-regime default.
   - `boundary` ⟶ emit warnings, no rank claims.
   - `backward_compat` ⟶ (`tumor_plus_differential_protection`, σ_floor = 0.05, δ = 0.2) — an explicit opt-in to V2.5-diff for users who need AUC parity with the pre-ag scored JSONLs, not a default.
4. **`p_diff` re-derivation under SE-on-mean variance** (§3.3) as a second probabilistic axis, so users can choose between the biological-overlap and the cohort-power interpretations.
5. **A continuous observation-confidence model** to replace the `EvidenceClass` discrete-bin scheme: a learned or calibrated confidence function over exact probe-distance, local probe density, mask quality, platform, and local concordance. This would address the central remaining coarseness of the §3.4 measurement model.
6. ~~Per-candidate moderated-t DMR baseline~~ **(done in this tag, §4.3 / §5.1 / §5.2.2):** limma-eBayes moderated-`t` on sample-level β matrices for GSE322563, GSE77348, GSE69914; probe-level first, then candidate-mapped via the V2.5 `observation.probe_id` field. AUC reported in §5.1 alongside Δβ-only / Δβ_z; WG cross-cohort panel in `examples/limma_cross_cohort_panel.md`.
7. **A second independent-lab MCF-7 / MCF-10A EPIC cohort**, if one becomes public, to establish reproducibility of the V2.5 claim at n ≥ 3 on matched cell-line pairs.
8. **Formal SCREEN cCRE Registry integration** when a non-challenged download path becomes available. The top-hit annotation pipeline already ships nearest-gene, CpG-island context, RepeatMasker overlap, and ENCODE DNase-HS cluster breadth (§5.5); DNase-HS is the v1 regulatory-activity proxy

for the formal cCRE Registry, which currently sits behind a JavaScript challenge on screen.wenglab.org and cannot be fetched directly.

---

7 · Methods — key equations

7.0 Canonical axis-naming table

With two V2.5-generation modes and four baselines, the paper uses the following canonical names; enum names are what `CohortConfig` accepts in YAML, prose names are what §5 tables and recommendations use:

Prose name	Code enum / field	Role	Recommendation
V1	`final_score`	stable-release default; `tie_band = 1` by construction	backward compatibility + deterministic top-K
V2	`probabilistic_mode: tumor_only`	first-pass composite; `p_prot` anti-predictive on tissue	diagnostic / audit only
V2.5-diff	`probabilistic_mode: tumor_plus_differential`	diagnostic V2.5; `p_targ × p_diff × p_trust`	retained for backward compat + AUC parity on cell-line cohorts
V2.5-sigmoid	`probabilistic_mode: tumor_plus_gap_sigmoid`	recommended V2.5 variant; `p_targ × p_gap_sigmoid × p_trust`	**recommended probabilistic discovery axis** (all tested non-boundary regimes)
Δβ-only	`naive_selectivity` (score-field)	raw methylation gap	honest baseline
Δβ_z	computed post-hoc from scored JSONL	σ-normalized raw gap	uncertainty-aware ablation baseline
limma-eBayes	probe-level Smyth 2004 eBayes	DMR comparator	methods-journal-standard baseline

The terms "probability-scale selectivity score" (paper-level) and the field `p_therapeutic_selectivity` (code-level) refer to the same emitted scalar under every probabilistic mode. "V2.5" without a `-diff` / `-sigmoid` qualifier refers to the *generation* (`tumor_plus_*`) and is used when a statement applies to both variants.

V1 `final_score`:

```
sequence_score      = PamFamily.weight
selectivity_score   = max(0, β_normal_mean − β_tumor_mean)
                    + max(0, β_normal_q25  − β_tumor_q75)
confidence_score    = EvidenceClass → {1.0, 0.7, 0.4, 0.1, 0.0}
heterogeneity_pen   = weight × max(0, β_tumor_IQR − threshold)
low_coverage_pen    = weight × max(0, (n_threshold − n_tumor) / n_threshold)
final_score         = sequence × selectivity × confidence
                    − heterogeneity_pen − low_coverage_pen
```

UNOBSERVED candidates are not penalized (verified by regression test).

V2 / V2.5 probabilistic factors:

```
p_targ = P(β_tumor < 0.30)   via method-of-moments Beta(α, β) fit,
                              fall back to piecewise-linear CDF when
                              σ² ≥ μ(1−μ). Incomplete beta via Lentz's
                              continued fraction, pure stdlib.
p_prot = P(β_normal ≥ 0.50)  via same CDF dispatch.     [V2 only]
p_diff = 1 − Φ((δ − (μ_n − μ_t)) / σ_Δ)
         where σ_Δ = √(max(σ_t, 0.05)² + max(σ_n, 0.05)²)
                           and σ_k = IQR_k / 1.349.      [V2.5-diff]
p_gap_sigmoid = 1 / (1 + exp(−((μ_n − μ_t) − δ) / σ_fixed))
         (δ = 0.2, σ_fixed = √2 × σ_floor ≈ 0.0707 by default;
          σ_fixed strictly positive, enforced at CohortConfig
          boundary). Numerically-stable closed form, pure stdlib.
                                                   [V2.5-sigmoid]
p_trust = EvidenceClass.base × min(1, min(n_t, n_n) / 30)
         (EvidenceClass.base ∈ {0.95, 0.75, 0.45, 0.15, 0.0}).
```

Composites by mode:

```
tumor_only                          : p_targ × p_trust
tumor_plus_normal_protection        : p_targ × p_prot × p_trust
tumor_plus_differential_protection  : p_targ × p_diff × p_trust
    (= V2.5-diff)
tumor_plus_gap_sigmoid              : p_targ × p_gap_sigmoid × p_trust
    (= V2.5-sigmoid; recommended probabilistic discovery axis, §5.2.2)
```

Validator enforces that the stored `p_therapeutic_selectivity` matches the product implied by the declared `mode` to within `math.isclose(rel_tol=1e-9, abs_tol=1e-12)`; a second validator enforces that the per-mode audit fields are populated iff the declared `mode` names them (`p_differential_protection` + `differential_delta` iff `tumor_plus_differential_protection`; `p_gap_sigmoid` + `sigma_fixed` + `differential_delta` iff `tumor_plus_gap_sigmoid`; none of the four iff `tumor_only` or `tumor_plus_normal_protection`).

Benchmark ranking contract. Let the scored set be N records, each with a score `s_i` and a ground-truth label `y_i ∈ {0, 1}` (1 for positive). The benchmark sorts records by the total order

```
(s_i, c_i)   with primary  -s_i  ascending
                  secondary c_i   ascending
```

where `c_i` is the candidate_id. Let `r_1, r_2, …, r_N` be the resulting ranking (smallest `(-s, c)` first). Define:

```
    top-K               := {r_1, …, r_K}
    cutoff_score        := s at position K in the sort, i.e. s_{r_K}
    tied_band           := {r_i : s_{r_i} = cutoff_score}          # records at the
cutoff
    tie_band_size_at_k  := |tied_band|
    k_band              := |tied_band ∩ top-K|                     # records in top-K at
the cutoff
    drop_slots          := tie_band_size_at_k − k_band             # records at cutoff
outside top-K
    band_pos            := |{r_i ∈ tied_band : y_{r_i} = 1}|
    committed_pos       := |{r_i ∈ top-K \ tied_band : y_{r_i} = 1}|
```

Then:

```
P@K_observed := (committed_pos + |{r_i ∈ tied_band ∩ top-K : y_{r_i} = 1}|) / K

P@K_min      := (committed_pos + max(0, band_pos − drop_slots)) / K
P@K_max      := (committed_pos + min(band_pos, k_band))          / K
```

`P@K_min` is the smallest possible value of P@K achievable by any tie-break policy that respects the primary score ordering; it is reached by pushing as many tied-band positives as possible outside the top-K (limited by `drop_slots` available). `P@K_max` is the symmetric upper bound: pull as many tied-band positives as possible into top-K (limited by `k_band` available). By construction:

```
P@K_min ≤ P@K_observed ≤ P@K_max
```

and the interval collapses to `{P@K_observed}` exactly when `tie_band_size_at_k = 1`.

The `R@K_min` / `R@K_max` counterparts replace the denominator with `n_positives`. `AUC` is the Mann-Whitney U / (n_pos · n_neg) with ties contributing 0.5 (standard tie-handling); it does not need an interval form because under the cid-ascending tie-break it is already invariant within tied score regions.

Data and code availability

- **Code**: https://github.com/AllisonH12/thermocas9. Cite tag **`memo-2026-04-22-al`** for this document. 245 tests pass under `uv run pytest -q`.
- **Citable archive (DOI)**: a Zenodo release archive of the tagged revision is planned at the time of preprint posting; the GitHub → Zenodo integration mints a DOI for each GitHub release tag. The DOI will be added to this section and to the citation block below before journal-version submission. Until then, the immutable git tag above is the canonical citable identifier.
- **Cohort data**: publicly-downloadable GEO series GSE322563, GSE77348, GSE69914, GSE68379; build scripts in `scripts/build_gse*_cohort.py` produce the per-probe summary TSVs in `data/derived/*_cohort/`. Positives-list builder at `scripts/build_roth_positives.py` (requires the Ensembl REST `/map` endpoint for the hg38 → hg19 liftover of Roth Fig. 5d coordinates).
- **Reference data**: UCSC hg19 `refGene.txt.gz` and `cpgIslandExt.txt.gz` (fetched on demand; gitignored).
- **Benchmark artifacts**: every `BenchmarkResult` JSONL row under `examples/*_roth_labels/` carries `precision_at_k`, `precision_at_k_{min,max}`, `recall_at_k`, `recall_at_k_{min,max}`, `roc_auc`, `tie_band_size_at_k`, and `tie_break_policy`.
- **Annotated top-20 TSVs**: under `examples/*/top20_annotated_*.tsv`.

Reproducing the cross-cohort matrix

```bash
# Given a clone of the repo:
uv venv && source .venv/bin/activate
uv pip install -e ".[dev]"
pytest                       # 245 tests, ~1 s

# Rebuild the catalog on chr5/6/10 (~2 min):
thermocas build-catalog \
    --reference data/raw/hg19/hg19_chr5_6_10.fa \
    --pam-model config/pam_model.yaml \
    --probe-annotation data/raw/probes_hg19.tsv \
    --probe-window-bp 500 \
    --output data/derived/catalog_hg19_chr5_6_10.jsonl

# Score + benchmark one cohort (~90 s per mode on a modern laptop):
```

```
thermocas score-cohort \
    --catalog data/derived/catalog_hg19_chr5_6_10.jsonl \
    --cohort data/derived/gse322563_differential.yaml \
    --pam-model config/pam_model.yaml \
    --backend summary \
    --probe-annotation data/derived/gse322563_cohort/probes.tsv \
    --tumor-summary  data/derived/gse322563_cohort/tumor_summary.tsv \
    --normal-summary data/derived/gse322563_cohort/normal_summary.tsv \
    --probabilistic \
    --output data/derived/scored_gse322563_differential.jsonl

thermocas benchmark \
    --scored data/derived/scored_gse322563_differential.jsonl \
    --positives data/derived/positives_roth_validated.txt \
    --cohort-name GSE322563-validated-differential \
    --score-field p_therapeutic_selectivity \
    --top-k 100 --no-enforce-holdout \
    --output bench.jsonl
```

`data/derived/scored_*.jsonl` files are not redistributed because they are 3.4 GB each and fully reproducible from the per-probe summary TSVs using the commands above.

Citation

If this paper or the associated code is useful in your work, please cite the upstream paper:

> Roth M.O., Shu Y., Zhao Y., Trasanidou D., Hoffman R.D., et al. Molecular basis for methylation-sensitive editing by Cas9. *Nature* (2026). DOI 10.1038/s41586-026-10384-z.

For the probabilistic scoring framework specifically, cite this paper plus the code URL above.

Acknowledgements

This work describes a framework built as an educational / research project. ThermoCas9′s biochemical characterization and the validated target sites are entirely due to Roth et al.; any errors in the re-implementation of the scoring or benchmarking are the author's.